

NEXORA TOUCH — API Integration Guide

v3

Version: 3.0 | **Updated:** 2026-06-08

Base: Clean Architecture .NET 8 API with CQRS (MediatR) and JWT authentication

Changes from v2: Corrected all endpoint payloads, response shapes, and error codes to match OpenAPI specification (specification.json). Updated pagination query param casing, staff search param, staff reorder field, dashboard response structures.

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Base URLs

Environment	Base URL	Swagger UI
Local	https://localhost:7012	https://localhost:7012/swagger
Test (Dev)	https://nexora-dev-api.vlinkhub.com	/api/index.html? url=/api/specification.json
Staging	<i>(chưa cập nhật)</i>	—
Production	<i>(chưa cập nhật)</i>	—

Authentication Header

All protected endpoints require:

```
Authorization: Bearer <accessToken>
```

Part 1 — API Reference

1. Authentication

Base route: POST /api/v1/authentication/*

Auth: AllowAnonymous (all endpoints)

POST /api/v1/authentication/signin

Sign in with email and password.

Rate limit: SignIn policy

Request body:

Field	Type	Required	Description
email	string	Yes	User email
password	string	Yes	User password

Response 200 :

```
{
  "accessToken": "eyJ...",
  "refreshToken": "abc...",
  "expiresIn": 3600
}
```

Response codes:

Status	ErrorCode	Description
200	—	Sign-in successful
400	USER_LOGIN_INVALID_USERNAME_OR_PASSWORD	Wrong credentials
400	USER_ACCOUNT_INACTIVE	Account suspended or inactive
429	COMMON_RATE_LIMIT_EXCEEDED	Too many sign-in attempts

POST /api/v1/authentication/signup

Register a new account.

Rate limit: SignUp policy

Request body:

Field	Type	Required	Description
email	string	Yes	Valid email address
password	string	Yes	Min 8 chars
confirmEmail	string	Yes	Must match email
type	string	Yes	"Merchant" or "User"
referralCode	string	No	Optional referral code

Response 201 :

```
{
  "isSuccess": true,
  "email": "user@example.com",
  "userId": "guid",
  "message": "Registration successful. Please verify your email.",
  "errorCode": null
}
```

Response codes:

Status	ErrorCode	Description
201	—	Account created; email verification sent
400	USER_EMAIL_ALREADY_EXISTS	Email taken
400	AUTH_PASSWORDS_D0_N0T_MATCH	Passwords don't match
400	USER_FEATURE_SIGNUP_DISABLED	Signup feature toggle disabled
429	COMMON_RATE_LIMIT_EXCEEDED	Rate limited

POST /api/v1/authentication/refresh-token

Exchange a refresh token for a new access token pair.

Rate limit: RefreshToken policy

Request body:

Field	Type	Required	Description
refreshToken	string	Yes	Current refresh token

Response 200 :

```
{
  "accessToken": "eyJ...",
  "refreshToken": "newRefresh...",
  "expiresIn": 3600
}
```

Response codes:

Status	ErrorCode	Description
200	—	New token pair issued
400	USER_INVALID_REFRESH_TOKEN	Token invalid or expired
401	AUTH_USER_NOT_AUTHENTICATED	Not authenticated

POST /api/v1/authentication/forgot-password

Request a password reset email.

Rate limit: ForgotPassword policy

Request body:

Field	Type	Required	Description
email	string	Yes	Registered email

Response 200 :

```
{
  "success": true
}
```

Always returns 200 to avoid email enumeration attacks.

POST /api/v1/authentication/reset-password

Reset password using the token from the email link.

Rate limit: ResetPassword policy

Request body:

Field	Type	Required	Description
token	string	Yes	Reset token from email
email	string	Yes	User email
newPassword	string	Yes	New password
confirmPassword	string	Yes	Must match newPassword

Response 200 :

```
{
  "success": true
}
```

Response codes:

Status	ErrorCode	Description
200	—	Password reset successful
400	USER_PASSWORD_RESET_TOKEN_EXPIRED	Token expired
400	AUTH_PASSWORDS_DO_NOT_MATCH	Passwords don't match

POST /api/v1/authentication/send-verification-email

Re-send the email verification link.

Rate limit: VerificationEmail policy

Request body:

Field	Type	Required	Description
email	string	Yes	Email to verify
ssoActivationToken	string	No	SSO activation token (required when verifying from SSO flow)

Response codes:

Status	ErrorCode	Description
200	—	Verification email sent
404	USER_NOT_FOUND	Email not registered
400	USER_EMAIL_ALREADY_VERIFIED	Already verified

POST /api/v1/authentication/verify-email

Confirm email with the token from the verification link.

Rate limit: VerificationEmail policy

Request body:

Field	Type	Required	Description
token	string	Yes	Email verification token
email	string	Yes	User email

Response 200 :

```
{
  "success": true,
  "message": "Email verified, account activated.",
  "redirectUrl": null
}
```

Response codes:

Status	ErrorCode	Description
200	—	Email verified, account activated
400	USER_INVALID_EMAIL_VERIFICATION_TOKEN	Token invalid
400	USER_EMAIL_VERIFICATION_TOKEN_EXPIRED	Token expired

POST /api/v1/authentication/passwordless-signin

Sign in via a one-time passwordless token (e.g., magic link from VlinkPay ecosystem).

Rate limit: PasswordlessSignIn policy

Request body:

Field	Type	Required	Description
token	string	Yes	Passwordless sign-in token
returnPath	string	No	Optional path to redirect after sign-in

Response 200 :

```
{
  "accessToken": "eyJ...",
  "refreshToken": "abc...",
  "expiresIn": 3600,
  "returnPath": "/dashboard"
}
```

Response codes:

Status	ErrorCode	Description
200	—	Sign-in successful
400	USER_INVALID_PASSWORDLESS_TOKEN	Token invalid or expired
404	USER_NOT_FOUND	User not found

2. User Profile

Base route: /api/v1/userprofile

Auth: [Authorize] (Bearer token required for all)

GET /api/v1/userprofile/me

Get the authenticated user's profile.

Response 200 :

```

{
  "id": "guid",
  "email": "user@example.com",
  "firstName": "John",
  "lastName": "Doe",
  "fullName": "John Doe",
  "phoneNumber": "+1...",
  "userType": "Merchant",
  "status": "Active",
  "profileImage": {
    "url": "https://...",
    "thumbnailUrl": "https://..."
  },
  "isEmailVerified": true,
  "isPhoneVerified": false,
  "isKYCVerified": false,
  "lastLoginAt": "2026-06-08T10:00:00Z",
  "dateOfBirth": null,
  "gender": null,
  "city": "Houston",
  "state": "TX",
  "country": "US",
  "zipCode": "77001",
  "address": "123 Main St",
  "contactInfo": null,
  "referralCode": "ABC123"
}

```

Response codes:

Status	ErrorCode	Description
200	—	Profile returned
401	COMMON_UNAUTHORIZED	Missing or invalid token
404	USER_NOT_FOUND	User not found

GET /api/v1/userprofile/verified-status

Check if the current user's email is verified.

Response 200 :

```
{
  "status": "Active",
  "profileType": "Merchant"
}
```

Response codes:

Status	ErrorCode	Description
200	—	Status returned
401	COMMON_UNAUTHORIZED	Missing or invalid token
404	USER_NOT_FOUND	User not found

GET /api/v1/userprofile/

Get a public profile by user ID (read-only public fields).

Path params: userId — UUID

Response 200 :

```
{
  "id": "guid",
  "firstName": "John",
  "lastName": "Doe",
  "fullName": "John Doe",
  "userType": "Merchant",
  "status": "Active",
  "profileImageUrl": "https://...",
  "bio": null,
  "city": "Houston",
  "state": "TX",
  "country": "US"
}
```

Response codes:

Status	ErrorCode	Description
200	—	Profile returned
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden
404	USER_NOT_FOUND	User not found

PUT /api/v1/userprofile/update

Update current user's profile.

Request body:

Field	Type	Required	Constraints
firstName	string	Yes	Max 100 chars
lastName	string	No	Max 100 chars
phoneNumber	string	No	International format
profileImageUrl	string	No	Valid URL, max 500 chars

Field	Type	Required	Constraints
city	string	No	Max 100 chars
state	string	No	Max 100 chars
country	string	No	Max 100 chars
zipCode	string	No	Max 20 chars
address	string	No	Max 200 chars
website	string	No	Valid URL
youtube	string	No	YouTube channel URL or handle
instagram	string	No	Instagram handle or URL
facebook	string	No	Facebook URL
twitter	string	No	Twitter/X handle or URL
tiktok	string	No	TikTok handle or URL

Returns updated `UserProfileResponseDto` (same shape as GET `/me` response).

Response codes:

Status	ErrorCode	Description
200	—	Profile updated; returns updated profile
400	USER_PROFILE_FIRST_NAME_REQUIRED	Missing first name
400	USER_PROFILE_PHONE_NUMBER_INVALID_FORMAT	Invalid phone
401	COMMON_UNAUTHORIZED	Missing or invalid token
404	USER_NOT_FOUND	User not found

POST /api/v1/userprofile/delete-account

Request account deletion. Triggers a soft-delete workflow.

Response: 200 OK

3. Merchant Business — Onboarding

Base route: /api/v1/merchant

Auth: [Authorize] — must be ProfileType.Merchant

POST /api/v1/merchant/business

Create the merchant's business profile (Step 1 of onboarding wizard).
Automatically creates a Starter subscription with a 30-day trial and pre-seeds all PayoutMethodType payment methods (inactive by default).

Request body:

Field	Type	Required	Constraints
name	string	Yes	3–100 chars
businessType	string	No	e.g., "Nail Salon", "Restaurant"
address	string	No	
city	string	No	
state	string	No	
phone	string	No	
timeZone	string	No	IANA timezone string
website	string	No	
description	string	No	Max 300 chars
logoUrl	string	No	Pre-uploaded via /api/v1/images/upload
customSlug	string	No	Lowercase, alphanumeric + hyphens, max 50 chars

Response 201 :

```
{
  "businessId": "guid",
  "slug": "bitcoin-nail-bar"
}
```

Response codes:

Status	ErrorCode	Description
201	—	Business created
400	BUSINESS_ALREADY_EXISTS	Merchant already has a business
400	BUSINESS_NAME_REQUIRED	Missing name
400	BUSINESS_INVALID_SLUG_FORMAT	Slug contains invalid characters
403	USER_NOT_MERCHANT	Caller is not a Merchant profile type
401	COMMON_UNAUTHORIZED	Missing or invalid token

Slug auto-generation: If `customSlug` is omitted, a unique slug is generated from `name` (lowercased, spaces → hyphens, special chars stripped). Collisions are resolved by appending `-2` , `-3` , etc.

GET /api/v1/merchant/business

Get the authenticated merchant's business profile.

Response 200 :

```
{
  "id": "guid",
  "name": "Bitcoin Nail Bar",
  "slug": "bitcoin-nail-bar",
  "businessType": "Nail Salon",
  "address": "123 Main St",
  "city": "Houston",
  "state": "TX",
  "phone": "+17131234567",
  "logoUrl": "https://...",
  "description": "...",
  "isPublic": false,
  "onboardingStep": 1,
  "googleReviewUrl": null,
  "yelpUrl": null,
  "facebookUrl": null,
  "feedbackEmail": null,
  "subscription": {
    "plan": "Starter",
    "status": "Trialing",
    "trialEndsAt": "2026-07-05T00:00:00Z"
  }
}
```

Response codes:

Status	ErrorCode	Description
200	—	Business found
404	BUSINESS_NOT_FOUND	No business exists for this merchant
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

GET /api/v1/merchant/business/check-slug

Real-time slug availability check (used during onboarding wizard typing).

Query params:

Param	Type	Required	Description
slug	string	No	Desired slug to check

Response 200 :

```
{
  "isAvailable": false,
  "suggestion": "bitcoin-nail-bar-2"
}
```

Response codes:

Status	ErrorCode	Description
401	COMMON_UNAUTHORIZED	Missing or invalid token

PUT /api/v1/merchant/business/review-links

Update external review platform links (Step 2 of onboarding wizard). All fields are optional.

Request body:

Field	Type	Required	Description
googleReviewUrl	string	No	Full Google Maps review URL
yelpUrl	string	No	Yelp business page URL
facebookUrl	string	No	Facebook page URL
feedbackEmail	string	No	Email for private negative feedback notifications

Response: 200 OK (no body)

Business rule BR-OB01: At least one review link is required for the review routing feature to work on the customer touch page. If none are set, ratings ≥ 4 will not redirect to external reviews.

Response codes:

Status	ErrorCode	Description
200	—	Updated
400	COMMON_VALIDATION_ERROR	Validation failed
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden
404	BUSINESS_NOT_FOUND	Business not found

PUT /api/v1/merchant/business/logo

Upload business logo. Accepts multipart form data. Logo is auto-resized to 400×400px and uploaded to S3.

Content-Type: multipart/form-data

Form fields:

Field	Type	Required	Description
logo	file	Yes	Image file (PNG/JPEG/WebP)

Response 200 :

```
{
  "logoUrl": "https://storage.nexora.vlinkpay.com/nexora/businesses/.../logo.png"
}
```

Response codes:

Status	ErrorCode	Description
200	—	Logo uploaded
400	BUSINESS_LOGO_UPLOAD_FAILED	Upload error
400	IMAGE_FILE_SIZE_EXCEEDED	File too large
400	IMAGE_UNSUPPORTED_FILE_TYPE	Invalid format

Status	ErrorCode	Description
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden
404	BUSINESS_NOT_FOUND	Business not found

POST /api/v1/merchant/business/complete-onboarding

Mark onboarding complete (Step 5). Sets `Business.IsPublic = true` and merchant's profile status to `Active`.

Request: No body.

Response: 200 OK

Response codes:

Status	ErrorCode	Description
200	—	Onboarding complete, business is now public
404	BUSINESS_NOT_FOUND	Business not created yet
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

4a. Merchant Staff Management

Base route: /api/v1/merchant/staff

Auth: [Authorize] — business owner only

POST /api/v1/merchant/staff/invite

Invite a new staff member (Luồng A). Sends a magic-link email if `invitedEmail` is provided. Token expires in **7 days**.

Request body:

Field	Type	Required	Constraints
<code>invitedName</code>	string	Yes	Max 100 chars
<code>invitedPhone</code>	string	Conditional	Required if <code>invitedEmail</code> is empty
<code>invitedEmail</code>	string	Conditional	Required if <code>invitedPhone</code> is empty
<code>invitedPosition</code>	string	No	Max 100 chars

Response 201 :

```
{
  "inviteId": "guid"
}
```

Response codes:

Status	ErrorCode	Description
201	—	Invite created, email sent
400	STAFF_PHONE_OR_EMAIL_REQUIRED	Neither phone nor email provided
400	STAFF_INVITED_NAME_REQUIRED	Name missing
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

POST /api/v1/merchant/staff/{inviteId}/resend

Resend an existing pending invite.

Path params: `inviteId` — UUID of the `StaffInvite`

Response: 204 No Content

Response codes:

Status	ErrorCode	Description
204	—	Invite resent
404	STAFF_INVITE_NOT_FOUND	Invite not found
400	STAFF_INVITE_NOT_PENDING	Invite already accepted or cancelled
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

POST /api/v1/merchant/staff/{inviteId}/resend-invite

Alternate endpoint to resend an existing pending invite (use interchangeably with /resend).

Path params: inviteId — UUID of the StaffInvite

Response: 204 No Content

Response codes:

Status	ErrorCode	Description
204	—	Invite resent
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden
404	STAFF_INVITE_NOT_FOUND	Invite not found
400	STAFF_INVITE_NOT_PENDING	Invite already accepted or cancelled

GET /api/v1/merchant/staff

Get the paginated staff list for the authenticated merchant's business. Includes both active links and pending invites.

Query params:

Param	Type	Required	Default	Description
Keyword	string	No	—	Case-insensitive search on DisplayName, StaffCode, email, phone
PageNumber	int	No	1	Page number (1-based)
PageSize	int	No	20	Items per page

Response 200 :

```

{
  "items": [
    {
      "id": "guid",
      "itemType": "link",
      "staffProfileId": "guid",
      "staffCode": "S12345678",
      "displayName": "Linh Nguyen",
      "photoUrl": "https://...",
      "position": "Nail Tech",
      "status": "Active",
      "sortOrder": 0,
      "isProfileComplete": true,
      "tipCount": 42,
      "averageRating": 4.8
    },
    {
      "id": "guid",
      "itemType": "invite",
      "staffProfileId": null,
      "staffCode": null,
      "displayName": "Maria Garcia",
      "photoUrl": null,
      "position": "Stylist",
      "status": "Pending",
      "sortOrder": 1,
      "isProfileComplete": false,
      "tipCount": 0,
      "averageRating": 0
    }
  ],
  "pageNumber": 1,
  "totalPages": 3,
  "totalCount": 42,
  "hasNextPage": true,
  "hasPreviousPage": false
}

```

itemType is "link" for accepted staff (has staffProfileId) or "invite" for pending invites (staffProfileId is null). Linked staff always appear before pending invites on the same page.

Response codes:

Status	ErrorCode	Description
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

GET /api/v1/merchant/staff/search

Search for an existing staff profile by phone or email (used for Luồng B — link existing staff).

Query params:

Param	Type	Required	Description
q	string	No	Search by phone number or email

Response 200 : Array of matching staff profiles (limited public fields — display name, position, photo).

Response codes:

Status	ErrorCode	Description
200	—	Matching staff profiles returned
400	COMMON_VALIDATION_ERROR	Validation error
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

PUT /api/v1/merchant/staff/{staffLinkId}/status

Set a staff member's link status to `Active` or `Inactive` .

`Inactive` hides the staff member from the customer touch page but preserves all data.

Path params: `staffLinkId` — UUID of the `BusinessStaffLink`

Request body:

Field	Type	Required	Values
staffLinkId	string (UUID)	Yes	Must match the path parameter {staffLinkId}
status	string	Yes	"Active" "Inactive"

Response: 204 No Content

Response codes:

Status	ErrorCode	Description
204	—	Status updated
400	STAFF_INVALID_STATUS	Invalid status value
404	STAFF_LINK_NOT_FOUND	Link not found
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

PUT /api/v1/merchant/staff/reorder

Update the display order of staff on the touch page (drag-and-drop).

Request body:

Field	Type	Required	Description
items	array	Yes	Array of { id: UUID, sortOrder: integer } objects in desired order

Response: 204 No Content

Response codes:

Status	ErrorCode	Description
204	—	Order updated
400	COMMON_VALIDATION_ERROR	Validation error

Status	ErrorCode	Description
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

DELETE /api/v1/merchant/staff/

Unlink a staff member from the business. Does not delete the staff profile or payment data.

Path params: staffLinkId — UUID

Response: 204 No Content

Response codes:

Status	ErrorCode	Description
204	—	Unlinked
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden
404	STAFF_LINK_NOT_FOUND	Link not found

POST /api/v1/merchant/staff/link-request/

Send a link request to an existing staff profile (Luồng B).

Path params: staffProfileId — UUID of the target StaffProfile

Response: 204 No Content

Response codes:

Status	ErrorCode	Description
204	—	Request sent

Status	ErrorCode	Description
400	STAFF_ALREADY_LINKED_TO_BUSINESS	Staff already linked to this business
404	STAFF_PROFILE_NOT_FOUND	Staff profile not found
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

4b. Staff Self-Setup (Public)

Base route: /api/v1/staff

Auth: Mixed — public invite endpoints are AllowAnonymous ; PUT /profile requires Bearer token

GET /api/v1/staff/invite/

Get invite metadata from a magic-link token (called when the staff member opens the invite link).

Auth: AllowAnonymous

Path params: token — URL-safe Base64 invite token

Response 200 :

```
{
  "invitedName": "Maria Garcia",
  "invitedPosition": "Stylist",
  "businessName": "Bitcoin Nail Bar"
}
```

Response codes:

Status	ErrorCode	Description
200	—	Invite found
400	STAFF_INVITE_EXPIRED	Token expired (7 days)

Status	ErrorCode	Description
400	STAFF_INVITE_ALREADY_ACCEPTED	Already accepted
410	COMMON_GONE	Token not found or permanently expired

POST /api/v1/staff/invite/{token}/accept

Accept an invite and set up the staff profile. Creates a `UserProfile` (if the email/phone isn't registered), a `StaffProfile` , and pre-seeds 5 payment methods (all inactive).

Auth: AllowAnonymous

Path params: token — invite token from URL

Request body:

Field	Type	Required	Constraints
token	string	Yes	Invite token (same as URL path parameter)
displayName	string	Yes	2–100 chars
position	string	No	Max 100 chars
bio	string	No	Max 200 chars
photoUrl	string	No	Pre-uploaded URL

Response: 204 No Content

Response codes:

Status	ErrorCode	Description
204	—	Invite accepted, profile created
400	STAFF_INVITE_EXPIRED	Token expired
400	STAFF_INVITE_ALREADY_ACCEPTED	Already accepted
400	STAFF_DISPLAY_NAME_REQUIRED	Display name missing

Status	ErrorCode	Description
401	COMMON_UNAUTHORIZED	Missing or invalid token

PUT /api/v1/staff/profile

Update the authenticated staff member's own profile.

Auth: [Authorize] — Bearer token required

Request body:

Field	Type	Required	Constraints
displayName	string	Yes	2–100 chars
position	string	No	Max 100 chars
bio	string	No	Max 200 chars
photoUrl	string	No	

Response: 204 No Content

Response codes:

Status	ErrorCode	Description
204	—	Profile updated
400	STAFF_DISPLAY_NAME_REQUIRED	Missing display name
400	STAFF_DISPLAY_NAME_T00_SHORT	Display name < 2 chars
404	STAFF_PROFILE_NOT_FOUND	No staff profile linked to this user
401	COMMON_UNAUTHORIZED	Missing or invalid token

5. Touch Points & QR

Base route: /api/v1/merchant/touchpoints

Auth: [Authorize]

POST /api/v1/merchant/touchpoints

Create a new touch point. Generates a 1000×1000px QR PNG and uploads it to S3 automatically.

Starter plan limit: Maximum 3 touch points.

Request body:

Field	Type	Required	Constraints
name	string	Yes	2–100 chars
type	string	Yes	"Table" "FrontDesk" "Receipt" "StaffCard"
assignedStaffProfileId	string (UUID)	No	Required when type = "StaffCard"

Response 201 :

```
{
  "touchPointId": "guid",
  "qrImageUrl": "https://storage.nexora.vlinkpay.com/nexora/touchpoints/.../table-1.png"
}
```

Touch point URL format: {baseUrl}/touch/{businessSlug}/{touchPointSlug}

e.g., https://nexora.vlinkpay.com/touch/bitcoin-nail-bar/table-1

Response codes:

Status	ErrorCode	Description
201	—	Touch point created

Status	ErrorCode	Description
400	TOUCHPOINT_STARTER_LIMIT_REACHED	Starter plan limit (3) reached
400	TOUCHPOINT_NAME_REQUIRED	Missing name
400	TOUCHPOINT_INVALID_TYPE	Invalid touch point type
403	USER_NOT_MERCHANT	Not a merchant
401	COMMON_UNAUTHORIZED	Missing or invalid token

GET /api/v1/merchant/touchpoints

List paginated active (non-deleted) touch points for the merchant's business.

Query params:

Param	Type	Required	Default	Description
PageNumber	int	No	1	Page number (1-based)
PageSize	int	No	20	Items per page
Name	string	No	—	Case-insensitive filter on touch point name

Response 200 :

```
{
  "items": [
    {
      "id": "guid",
      "name": "Table 1",
      "slug": "table-1",
      "type": "Table",
      "url": "https://nexora.vlinkpay.com/touch/bitcoin-nail-bar/table-1",
      "qrImageUrl": "https://...",
      "assignedStaffProfileId": null,
      "isActive": true,
      "createdAt": "2026-06-01T12:00:00Z"
    }
  ],
  "pageNumber": 1,
  "totalPages": 2,
  "totalCount": 25,
  "hasNextPage": true,
  "hasPreviousPage": false
}
```

Sorted by createdAt DESC (newest first).

Response codes:

Status	ErrorCode	Description
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

GET /api/v1/merchant/touchpoints/{id}/download

Download the QR code as PNG or PDF.

Path params: id — UUID

Query params:

Param	Type	Required	Values
format	string	No	"png" "pdf"

Response:

- format=png → image/png — 1000×1000px
- format=pdf → application/pdf — A4 with QR + touch point name + Nexora branding

Response codes:

Status	ErrorCode	Description
200	—	File returned as download
400	TOUCHPOINT_INVALID_DOWNLOAD_FORMAT	Invalid format value
404	TOUCHPOINT_NOT_FOUND	Touch point not found
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

DELETE /api/v1/merchant/touchpoints/

Soft-delete a touch point. Sets DeletedAt — preserves analytics history.

Path params: id — UUID

Response: 204 No Content

Response codes:

Status	ErrorCode	Description
204	—	Deleted
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden
404	TOUCHPOINT_NOT_FOUND	Touch point not found

6. Staff Payment Methods

Base route: /api/v1/staff/payment-methods

Auth: [Authorize] — staff member only (ownership enforced)

Privacy rule BR-ST02: Payment account info (CashApp tag, Venmo handle, Zelle phone/email, PayPal link) belongs exclusively to the staff member. Merchant endpoints **cannot** read raw `accountInfo` values.

Staff profiles are pre-seeded with 5 payment method records (inactive) upon accepting an invite.

GET /api/v1/staff/payment-methods

Get the authenticated staff member's own payment methods.

Response 200 :

```
[
  {
    "id": "guid",
    "type": "Zelle",
    "accountInfo": "+17131234567",
    "imageUrl": null,
    "isActive": true,
    "isConfigured": true
  },
  {
    "id": "guid",
    "type": "CashApp",
    "accountInfo": null,
    "imageUrl": null,
    "isActive": false,
    "isConfigured": false
  }
]
```

`isConfigured = true` when `accountInfo` is set. A method must be both `isActive` and `isConfigured` to appear on the customer touch page.

Response codes:

Status	ErrorCode	Description
200	—	Payment methods returned
401	COMMON_UNAUTHORIZED	Missing or invalid token
404	STAFF_PROFILE_NOT_FOUND	Staff profile not found

PUT /api/v1/staff/payment-methods/

Update a payment method's account info and optional custom image. This is a **full replace** — sending `null` for a field clears it. After saving, the system recalculates `StaffProfile.IsProfileComplete` (true when at least one method has `isActive = true` and a non-empty `accountInfo`).

Path params: `id` — UUID of the `StaffPaymentMethod`

Request body:

Field	Type	Required	Constraints
<code>accountInfo</code>	<code>string</code> <code>null</code>	No	Max 255 chars. Value depends on type: phone/email for Zelle, <code>\$tag</code> for CashApp, <code>@handle</code> for Venmo, PayPal.me URL for PayPal. Sending <code>null</code> clears the field.
<code>imageUrl</code>	<code>string</code> <code>null</code>	No	Max 500 chars. Custom QR/avatar URL. Sending <code>null</code> clears the field.

Response 200 :

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "type": "Zelle",
  "accountInfo": "+17131234567",
  "imageUrl": null,
  "isActive": false,
  "isConfigured": true
}
```

`isConfigured` is `true` when `accountInfo` is non-empty. It is computed — not a stored field.

Response codes:

Status	ErrorCode	Description
200	—	Updated successfully
400	STAFF_ACCOUNT_INFO_TOO_LONG	<code>accountInfo</code> exceeds 255 chars
400	STAFF_IMAGE_URL_TOO_LONG	<code>imageUrl</code> exceeds 500 chars
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	STAFF_PAYMENT_METHOD_ACCESS_DENIED	Attempting to update another user's method
404	STAFF_PAYMENT_METHOD_NOT_FOUND	Method not found

PATCH /api/v1/staff/payment-methods/{id}/toggle

Toggle a payment method on (`isActive = true`) or off (`isActive = false`).

Path params: `id` — UUID

No request body.

Response 200 : Updated `StaffPaymentMethodDto` .

Business rule BR-ST01: A staff member must have at least one active AND configured payment method to be visible on the customer touch page (`isProfileComplete = true`).

Response codes:

Status	ErrorCode	Description
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	STAFF_PAYMENT_METHOD_ACCESS_DENIED	Attempting to toggle another user's method
404	STAFF_PAYMENT_METHOD_NOT_FOUND	Method not found

7. Merchant Payment Methods

Base route: /api/v1/merchant/payment-methods

Auth: [Authorize] — merchant only

Merchant payment methods (`BusinessPaymentMethod`) are used in the **multi-staff tip flow** (Section 9). Customers tip through the business's payment method (e.g., the business's Venmo) and the business distributes to staff. This is separate from individual staff payment methods. All `PayoutMethodType` methods are pre-seeded on business creation (all inactive).

GET /api/v1/merchant/payment-methods

Get the merchant's business payment methods.

Response 200 :

```
[
  {
    "id": "guid",
    "type": "Venmo",
    "accountInfo": "@bitcoin-nail-bar",
    "imageUrl": null,
    "isActive": true,
    "isConfigured": true,
    "businessKybStatus": null
  },
  {
    "id": "guid",
    "type": "VlinkPay",
    "accountInfo": null,
    "imageUrl": null,
    "isActive": false,
    "isConfigured": false,
    "businessKybStatus": "NotStarted"
  }
]
```

`VlinkPay` type requires KYC/KYB approval before it can be toggled active.

Response codes:

Status	ErrorCode	Description
200	—	Payment methods returned
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

PUT /api/v1/merchant/payment-methods/

Update a business payment method's account info and optional image. This is a **full replace** — sending `null` for a field clears it.

Path params: `id` — UUID of the `BusinessPaymentMethod`

Request body:

Field	Type	Required	Constraints
<code>accountInfo</code>	<code>string</code> <code>null</code>	No	Max 255 chars. Typically the business's handle or phone for the given payment type. Sending <code>null</code> clears the field.
<code>imageUrl</code>	<code>string</code> <code>null</code>	No	Max 500 chars. Sending <code>null</code> clears the field.

Response 200 :

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "type": "Venmo",
  "accountInfo": "@bitcoin-nail-bar",
  "imageUrl": null,
  "isActive": false,
  "isConfigured": true,
  "businessKybStatus": null
}
```

`isConfigured` is `true` when `accountInfo` is non-empty. `businessKybStatus` is non-null only for the `VlinkPay` type.

Response codes:

Status	ErrorCode	Description
200	—	Updated successfully
400	BUSINESS_PAYMENT_METHOD_ACCOUNT_INFO_TOO_LONG	<code>accountInfo</code> exceeds 255 chars
400	BUSINESS_PAYMENT_METHOD_IMAGE_URL_TOO_LONG	<code>imageUrl</code> exceeds 500 chars
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	BUSINESS_PAYMENT_METHOD_ACCESS_DENIED	Method belongs to a different merchant
403	USER_NOT_MERCHANT	Caller is not a Merchant profile type
404	BUSINESS_PAYMENT_METHOD_NOT_FOUND	Method not found

PATCH /api/v1/merchant/payment-methods/{id}/toggle

Toggle a business payment method on/off.

Path params: `id` — UUID

No request body.

Response 200 : Updated `BusinessPaymentMethodDto` .

Response codes:

Status	ErrorCode	Description
200	—	Toggled
400	BUSINESS_PAYMENT_METHOD_CANNOT_TOGGLE_VLINKPAY	VlinkPay requires KYC first
400	BUSINESS_PAYMENT_METHOD_VLINKPAY_KYC_REQUIRED	KYC not approved

Status	ErrorCode	Description
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	BUSINESS_PAYMENT_METHOD_ACCESS_DENIED	Forbidden
404	BUSINESS_PAYMENT_METHOD_NOT_FOUND	Method not found

8. Customer Touch Page (Public)

Base route: /api/v1/touch

Auth: AllowAnonymous (all endpoints)

Rate limit: PublicTouchPolicy

These are the guest-facing endpoints called from the customer's browser when they scan a QR code.

GET /api/v1/touch/{businessSlug}/

Load the touch page data (business info, touch point info, staff list).

The `sessionId` query param is used for analytics tracking — generate a UUID on the frontend per session.

Path params:

Param	Description
businessSlug	Business URL slug (e.g., bitcoin-nail-bar)
touchPointSlug	Touch point URL slug (e.g., table-1)

Query params:

Param	Type	Required	Description
sessionId	string (UUID)	No	Session identifier for analytics grouping

Response 200 :


```

{
  "business": {
    "name": "Bitcoin Nail Bar",
    "logoUrl": "https://...",
    "description": "Premium nail salon..."
  },
  "touchPoint": {
    "id": "guid",
    "name": "Table 1",
    "type": "Table",
    "assignedStaffProfileId": null
  },
  "staff": [
    {
      "id": "guid",
      "displayName": "Linh Nguyen",
      "photoUrl": "https://...",
      "position": "Nail Tech",
      "bio": "5 years experience",
      "availablePaymentMethods": ["CashApp", "Venmo"]
    }
  ]
}

```

Privacy: availablePaymentMethods lists method **type names only** — raw accountInfo values are never exposed.

StaffCard: When touchPoint.type = "StaffCard", staff will contain exactly one entry (the assigned staff member).

Inactive touch point: Returns 404 — redirected to "unavailable" page on the frontend.

Response codes:

Status	ErrorCode	Description
200	—	Touch page data loaded
404	TOUCHPOINT_NOT_FOUND	Slug doesn't exist or touch point is inactive

GET /api/v1/touch/payment-link

Get the payment deep link (or Zelle copy info) for the selected staff, method, and amount.
Called when the customer taps "Pay" to redirect to the payment app.

Query params:

Param	Type	Required	Description
staffId	UUID	Yes	StaffProfile.Id
method	integer	Yes	PaymentMethod enum value (see Enums section)
amount	decimal	Yes	Tip amount (1.00–500.00)

Response 200 :

```
{
  "redirectUrl": "https://cash.app/$linh-nguyen/25.00",
  "zellePhone": null,
  "zelleEmail": null,
  "appleCashPhone": null
}
```

For **Zelle**: redirectUrl is null; zellePhone or zelleEmail is populated for the frontend to display a copy-paste UI.

For **Apple Cash**: redirectUrl is null; appleCashPhone is populated.

For **CashApp, Venmo, PayPal**: redirectUrl is the deep link.

Response codes:

Status	ErrorCode	Description
200	—	Payment link returned
400	TIP_INVALID_PAYMENT_METHOD	Invalid method value
400	TIP_PAYMENT_METHOD_NOT_CONFIGURED	Staff hasn't set up this payment method
404	STAFF_PROFILE_NOT_FOUND	Staff not found

POST /api/v1/touch/tip

Create a single-staff tip record with status `Initiated` . Call this when the customer taps "Pay" (after payment link is shown). Records the intent to tip and logs an analytics event.

Request body:

Field	Type	Required	Constraints
touchPointId	UUID	Yes	Active touch point
staffProfileId	UUID	No	Null if skipping staff selection
amount	decimal	Yes	1.00–500.00
paymentMethod	integer	Yes	PaymentMethod enum value
sessionId	string	No	Session UUID for analytics

Response 201 :

```
{
  "tipId": "guid"
}
```

Response codes:

Status	ErrorCode	Description
201	—	Tip initiated
400	TIP_AMOUNT_T00_LOW	Amount < \$1.00
400	TIP_AMOUNT_T00_HIGH	Amount > \$500.00
400	TIP_INVALID_PAYMENT_METHOD	Invalid method enum
404	TOUCHPOINT_NOT_FOUND	Touch point not found or inactive

POST /api/v1/touch/tip/skip

Record that the customer chose to skip the tip (used for analytics and to create a Skipped tip record before the review step).

Request body:

Field	Type	Required	Description
touchPointId	UUID	Yes	Active touch point
staffProfileId	UUID	No	Selected staff (if any)
sessionId	string	No	Session UUID

Response: 201 Created (no body)

POST /api/v1/touch/tip/{tipId}/confirm

Confirm a single-staff tip — called when the customer taps "Yes, I paid" on the confirmation screen. Updates tip status to Confirmed .

Path params: tipId — UUID from the POST /tip response

No request body.

Response: 200 OK

Response codes:

Status	ErrorCode	Description
200	—	Tip confirmed
400	TIP_ALREADY_CONFIRMED	Tip already in Confirmed state
404	TIP_NOT_FOUND	Tip not found

POST /api/v1/touch/review

Submit a customer review. Rating determines routing:

- **4–5 stars** → RoutingType = Public — frontend shows Google/Yelp buttons
- **1–3 stars** → RoutingType = Private — private feedback collected; email sent to merchant within 5 minutes

Request body:

Field	Type	Required	Constraints
touchPointId	UUID	Yes	
tipId	UUID	No	Link to a prior tip if applicable
staffProfileId	UUID	No	Staff being reviewed
rating	integer	Yes	1–5
comment	string	No	Max 500 chars
customerEmail	string	No	Valid email, max 200 chars (voluntary, for private feedback reply)
customerName	string	No	Max 100 chars

Response 201 :

```
{
  "reviewId": "guid"
}
```

The frontend should check `rating >= 4` locally to decide whether to show "Leave a Google/Yelp review" buttons before calling this endpoint, as the routing type is determined server-side based on the same threshold.

Response codes:

Status	ErrorCode	Description
201	—	Review created
400	REVIEW_RATING_RANGE	Rating not between 1–5

Status	ErrorCode	Description
400	REVIEW_COMMENT_MAX_LENGTH	Comment > 500 chars
400	REVIEW_CUSTOMER_EMAIL_INVALID	Invalid email format
404	TOUCHPOINT_NOT_FOUND	Touch point not found

POST /api/v1/touch/review/{reviewId}/track-google

Log that the customer clicked the Google review button. Records `GoogleClickedAt` timestamp.

Path params: `reviewId` — UUID

No request body.

Response: 200 OK

POST /api/v1/touch/review/{reviewId}/track-yelp

Log that the customer clicked the Yelp review button. Records `YelpClickedAt` timestamp.

Path params: `reviewId` — UUID

No request body.

Response: 200 OK

9. Multi-Staff Tip (Public)

Base route: `/api/v1/tips`

Auth: `AllowAnonymous`

Rate limit: `PublicTouchPolicy`

The multi-staff tip flow allows a customer to tip **two or more staff members** in a single transaction through the **business's payment method** (e.g., the salon's Venmo). The business then distributes the

individual amounts to each staff member.

POST /api/v1/tips/multi-staff

Create a multi-staff tip. Validates that:

- At least **2 distinct** staff profile IDs are provided
- All staff are active at the given business
- The selected `businessPaymentMethodId` is active and configured
- Each individual `amount` $\geq$ \$1.00
- Total sum of all `amount` values $\leq$ \$500.00

Uses a database transaction — rolls back entirely on any error.

Request body:

Field	Type	Required	Constraints
<code>businessId</code>	UUID	Yes	Business making the transaction
<code>touchPointId</code>	UUID	Yes	Active touch point
<code>businessPaymentMethodId</code>	UUID	Yes	Active, configured <code>BusinessPaymentMethod.Id</code>
<code>tipItems</code>	array	Yes	Minimum 2 items; no duplicate <code>staffProfileId</code>
<code>tipItems[].staffProfileId</code>	UUID	Yes	Must be active at the business
<code>tipItems[].amount</code>	decimal	Yes	≥ 1.00 ; total sum ≤ 500.00

Example request:

```

{
  "businessId": "guid",
  "touchPointId": "guid",
  "businessPaymentMethodId": "guid",
  "tipItems": [
    { "staffProfileId": "guid-linh", "amount": 15.00 },
    { "staffProfileId": "guid-maria", "amount": 10.00 }
  ]
}

```

Response 201 :

```

{
  "tipId": "guid",
  "totalAmount": 25.00,
  "paymentMethodType": "Venmo",
  "tipItems": [
    { "staffProfileId": "guid-linh", "amount": 15.00 },
    { "staffProfileId": "guid-maria", "amount": 10.00 }
  ]
}

```

Response codes:

Status	ErrorCode	Description
201	—	Multi-staff tip created
400	TIP_BUSINESS_PAYMENT_METHOD_REQUIRED	businessPaymentMethodId missing or empty
400	TIP_MINIMUM_STAFF_COUNT	Fewer than 2 staff in tipItems
400	TIP_DUPLICATE_STAFF_PROFILE_ID	Same staffProfileId appears more than once
400	TIP_AMOUNT_TOO_LOW	An individual item amount < \$1.00
400	TIP_AMOUNT_TOO_HIGH	Total sum > \$500.00
400	TIP_INVALID_BUSINESS_PAYMENT_METHOD	Payment method not active/configured

Status	ErrorCode	Description
400	TIP_STAFF_NOT_ACTIVE_AT_BUSINESS	One or more staff not active at this business
404	TOUCHPOINT_NOT_FOUND	Touch point not found or inactive

PATCH /api/v1/tips/{id}/confirm

Confirm a multi-staff tip (customer tapped "Yes, I paid"). Updates `Tip.Status` to `Confirmed`.

Path params: `id` — UUID of the tip

No request body.

Response: 200 OK

Response codes:

Status	ErrorCode	Description
200	—	Confirmed
400	TIP_ALREADY_CONFIRMED	Already confirmed
404	TIP_NOT_FOUND	Tip not found

10. Public Business Payment Methods

Base route: `/api/v1/public/businesses`

Auth: `AllowAnonymous`

Rate limit: `PublicTouchPolicy`

Used during the multi-staff tip flow to display which payment methods the business has active, so the customer can choose how to pay.

GET /api/v1/public/businesses/{businessId}/payment-methods

Get all active payment methods for a business (visible to guests).

Path params: businessId — UUID

Response 200 :

```
[
  {
    "id": "guid",
    "type": "Venmo",
    "accountInfo": "@bitcoin-nail-bar",
    "imageUrl": null
  }
]
```

Only methods where IsActive = true and AccountInfo != null are returned.

GET /api/v1/public/businesses/{businessId}/payment-methods/

Get a single active payment method detail.

Path params: businessId — UUID, id — UUID

Response 200 : Single PublicPaymentMethodDto (same shape as list item).

Response codes:

Status	ErrorCode	Description
200	—	Found
404	BUSINESS_PAYMENT_METHOD_NOT_FOUND	Not found or inactive

11. Owner Dashboard

Base route: /api/v1/merchant/dashboard

Auth: [Authorize]

GET /api/v1/merchant/dashboard/overview

Get aggregate metrics (tips, scans, reviews) for the merchant's business within a date range.

Query params:

Param	Type	Required	Default
dateFrom	ISO 8601 datetime	No	30 days ago
dateTo	ISO 8601 datetime	No	Now

Response 200 :

```
{
  "tipsSummary": {
    "totalAmount": 1250.00,
    "totalCount": 48,
    "avgAmount": 26.04,
    "previousPeriodComparison": null
  },
  "scansSummary": {
    "totalPageViews": 312,
    "conversionRate": 0.154
  },
  "reviewsSummary": {
    "totalCount": 35,
    "avgRating": 4.6,
    "count4To5Stars": 28,
    "count1To3Stars": 7,
    "googleClickCount": 18,
    "yelpClickCount": 5
  }
}
```

GET /api/v1/merchant/dashboard/reviews

Get paginated reviews for the merchant's business. Includes both public and private reviews.

Query params:

Param	Type	Description
pageNumber	int	Default: 1
pageSize	int	Default: 10, max 50
routingType	string	Filter: "Public" "Private" (all if omitted)
isResolved	bool	Filter resolved/unresolved (for private feedback inbox)

Response 200 :

```
{
  "items": [
    {
      "id": "guid",
      "rating": 2,
      "comment": "Long wait time...",
      "routingType": "Private",
      "staffName": "Linh Nguyen",
      "touchPointName": "Table 1",
      "customerEmail": null,
      "customerName": null,
      "isResolved": false,
      "googleClickedAt": null,
      "yelpClickedAt": null,
      "createdAt": "2026-06-05T10:00:00Z"
    }
  ],
  "totalCount": 7,
  "pageNumber": 1,
  "totalPages": 1,
  "hasPreviousPage": false,
  "hasNextPage": false
}
```

Privacy: `customerEmail` and `customerName` are only present when voluntarily provided by the customer.

PUT /api/v1/merchant/dashboard/reviews/{id}/resolve

Mark a private review as resolved (feedback addressed).

Path params: `id` — UUID

No request body.

Response: 204 No Content

GET /api/v1/merchant/dashboard/staff

Get staff metrics for the date range.

Query params: `dateFrom` , `dateTo` (optional, same as overview)

Response 200 :

```
[
  {
    "staffProfileId": "guid",
    "displayName": "Linh Nguyen",
    "photoUrl": "https://...",
    "position": "Nail Tech",
    "tipCount": 30,
    "tipTotal": 750.00,
    "avgTip": 25.00,
    "avgRating": 4.9,
    "reviewCount": 12,
    "selectionCount": 95
  }
]
```

GET /api/v1/merchant/dashboard/touchpoints

Get touch point metrics for the date range.

Query params: dateFrom , dateTo (optional)

Response 200 :

```
[
  {
    "touchPointId": "guid",
    "name": "Table 1",
    "type": "Table",
    "scanCount": 150,
    "tipCount": 22,
    "tipTotal": 550.00,
    "ctr": 0.147,
    "avgRating": 4.5
  }
]
```

GET /api/v1/merchant/dashboard/tips-chart

Get time-series tip data for chart visualization within a date range. Returns one data point per day, zero-filled for days with no confirmed tips.

Query params:

Param	Type	Required	Description
DateFrom	ISO 8601 datetime	Yes	Start of date range (inclusive)
DateTo	ISO 8601 datetime	Yes	End of date range (inclusive); server extends to end-of-day automatically

Response 200 :

```

{
  "data": [
    {
      "date": "2026-06-01",
      "totalAmount": 250.00,
      "tipCount": 10,
      "avgAmount": 25.00
    },
    {
      "date": "2026-06-02",
      "totalAmount": 0,
      "tipCount": 0,
      "avgAmount": 0
    },
    {
      "date": "2026-06-03",
      "totalAmount": 180.00,
      "tipCount": 7,
      "avgAmount": 25.71
    }
  ]
}

```

- date format is yyyy-MM-dd (DateOnly). Results are sorted ascending by date.
- Only tips with Status = Confirmed are counted.
- Days with no confirmed tips are included with totalAmount = 0 , tipCount = 0 , avgAmount = 0 (zero-fill).
- avgAmount = 0 when tipCount = 0 — no division by zero.
- DateTo is extended to end-of-day server-side — pass the calendar date directly (e.g., 2026-06-08).

Response codes:

Status	ErrorCode	Description
200	—	Chart data returned
400	COMMON_VALIDATION_ERROR	DateFrom or DateTo missing, or DateTo < DateFrom
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	COMMON_FORBIDDEN	Forbidden

Status	ErrorCode	Description
404	BUSINESS_NOT_FOUND	No business for this merchant

12. Image Upload

⚠ **Note:** These endpoints are not currently included in the OpenAPI specification (`specification.json`) but are implemented in the API. The documentation below is based on source code. Verify with Swagger UI if in doubt.

Base route: `/api/v1/images`

POST `/api/v1/images/upload`

Upload a single image (authenticated).

Auth: [Authorize]

Rate limit: ImageUpload

Content-Type: `multipart/form-data`

Form fields:

Field	Type	Required	Constraints
file	file	Yes	PNG/JPEG/WebP; max 10 MB

Response 200 :

```
{
  "fileUrl": "https://storage.nexora.vlinkpay.com/nexora/images/..."
}
```

Response codes:

Status	ErrorCode	Description
200	—	Uploaded
400	IMAGE_FILE_REQUIRED	No file provided
400	IMAGE_UNSUPPORTED_FILE_TYPE	Not PNG/JPEG/WebP
400	IMAGE_FILE_SIZE_EXCEEDED	> 10MB
400	IMAGE_UPLOAD_FAILED	S3 upload error

POST /api/v1/images/public/upload

Upload a single image without authentication (used on the staff invite accept page).

Auth: AllowAnonymous

Rate limit: PublicImageUpload

Same request/response as /images/upload .

13. Notifications

⚠ **Note:** These endpoints are not currently included in the OpenAPI specification (specification.json) but are implemented in the API. The documentation below is based on source code. Verify with Swagger UI if in doubt.

Base route: /api/v1/notifications

Auth: [Authorize]

GET /api/v1/notifications

Get paginated notifications for the current user.

Query params: pageNumber , pageSize

Response 200 : Paginated list of notifications.

GET /api/v1/notifications/unread-count

Get the number of unread notifications.

Response 200 :

```
{  
  "count": 3  
}
```

PUT /api/v1/notifications/{id}/read

Mark a notification as read.

Path params: id — UUID

Response: 200 OK

PUT /api/v1/notifications/read-all

Mark all notifications as read.

Response: 200 OK

14. Tip Transaction History (Merchant)

Base route: /api/v1/merchant/dashboard/tips

Auth: [Authorize] — merchant only

Returns a paginated list of individual tip transactions for the authenticated merchant's business.
Supports multiple filters for searching and reconciliation.

GET /api/v1/merchant/dashboard/tips

Query params:

Param	Type	Required	Default	Description
PageNumber	int	No	1	Page number
PageSize	int	No	20	Items per page
DateFrom	ISO 8601 datetime	No	—	Filter from date (inclusive)
DateTo	ISO 8601 datetime	No	—	Filter to end of day (inclusive)
Status	int	No	all	TipStatus enum: 0 =Initiated, 1 =Confirmed, 2 =Skipped
PaymentMethod	int	No	all	PaymentMethod enum (see Enums section)
StaffProfileId	UUID	No	—	Exact staff match (single-staff and multi-staff tips)
StaffSearch	string	No	—	Free-text search by StaffCode / DisplayName / Email / Phone. Ignored when StaffProfileId is set
TouchPointId	UUID	No	—	Filter by touch point
IsMultiStaff	bool	No	all	true = multi-staff only, false = single-staff only

StaffSearch behavior: Case-insensitive Contains match against StaffCode , DisplayName , email, and phone number for both single-staff tips (direct staff) and multi-staff tips (any TipItem staff).

Response 200 :

```

{
  "items": [
    {
      "id": "guid",
      "amount": 25.00,
      "status": 1,
      "paymentMethod": 1,
      "isMultiStaff": true,
      "touchPointId": "guid",
      "touchPointName": "Table 1",
      "staffProfileId": null,
      "staffName": null,
      "tipItems": [
        { "staffProfileId": "guid-linh", "staffName": "Linh Nguyen", "amount": 15.00 },
        { "staffProfileId": "guid-maria", "staffName": "Maria Garcia", "amount": 10.00 }
      ],
      "createdAt": "2026-06-05T10:00:00Z",
      "confirmedAt": "2026-06-05T10:02:00Z"
    },
    {
      "id": "guid",
      "amount": 20.00,
      "status": 1,
      "paymentMethod": 0,
      "isMultiStaff": false,
      "touchPointId": "guid",
      "touchPointName": "Front Desk",
      "staffProfileId": "guid-linh",
      "staffName": "Linh Nguyen",
      "tipItems": [],
      "createdAt": "2026-06-05T09:00:00Z",
      "confirmedAt": "2026-06-05T09:01:00Z"
    }
  ],
  "totalCount": 48,
  "pageNumber": 1,
  "totalPages": 3
}

```

`tipItems` is an empty array for single-staff tips; `staffProfileId` / `staffName` are null for multi-staff tips.

confirmedAt is null when status is Initiated or Skipped .

Response codes:

Status	ErrorCode	Description
200	—	Tip list returned
401	COMMON_UNAUTHORIZED	Missing or invalid token
403	USER_NOT_MERCHANT	Caller is not a Merchant profile type
404	BUSINESS_NOT_FOUND	No business exists for this merchant

15. Staff Tip History

Base route: /api/v1/staff/tips

Auth: [Authorize] — staff member only

Returns a paginated list of tips received by the authenticated staff member. For multi-staff tips, amount reflects only the staff member's individual portion.

GET /api/v1/staff/tips

Query params:

Param	Type	Required	Default	Description
PageNumber	int	No	1	Page number
PageSize	int	No	20	Items per page
DateFrom	ISO 8601 datetime	No	—	Filter from date (inclusive)
DateTo	ISO 8601 datetime	No	—	Filter to end of day (inclusive)

Param	Type	Required	Default	Description
Status	int	No	all	TipStatus enum: 0 =Initiated, 1 =Confirmed, 2 =Skipped

Response 200 :

```
{
  "items": [
    {
      "id": "guid",
      "amount": 15.00,
      "totalAmount": 25.00,
      "status": 1,
      "paymentMethod": 1,
      "isMultiStaff": true,
      "touchPointName": "Table 1",
      "businessName": "Bitcoin Nail Bar",
      "createdAt": "2026-06-05T10:00:00Z",
      "confirmedAt": "2026-06-05T10:02:00Z"
    },
    {
      "id": "guid",
      "amount": 20.00,
      "totalAmount": 20.00,
      "status": 1,
      "paymentMethod": 0,
      "isMultiStaff": false,
      "touchPointName": "Front Desk",
      "businessName": "Bitcoin Nail Bar",
      "createdAt": "2026-06-05T09:00:00Z",
      "confirmedAt": "2026-06-05T09:01:00Z"
    }
  ],
  "totalCount": 30,
  "pageNumber": 1,
  "totalPages": 2
}
```

For multi-staff tips: `amount` = this staff member's portion (from `TipItem.Amount`); `totalAmount` = the full tip the customer paid.

For single-staff tips: `amount` = `totalAmount` .

Response codes:

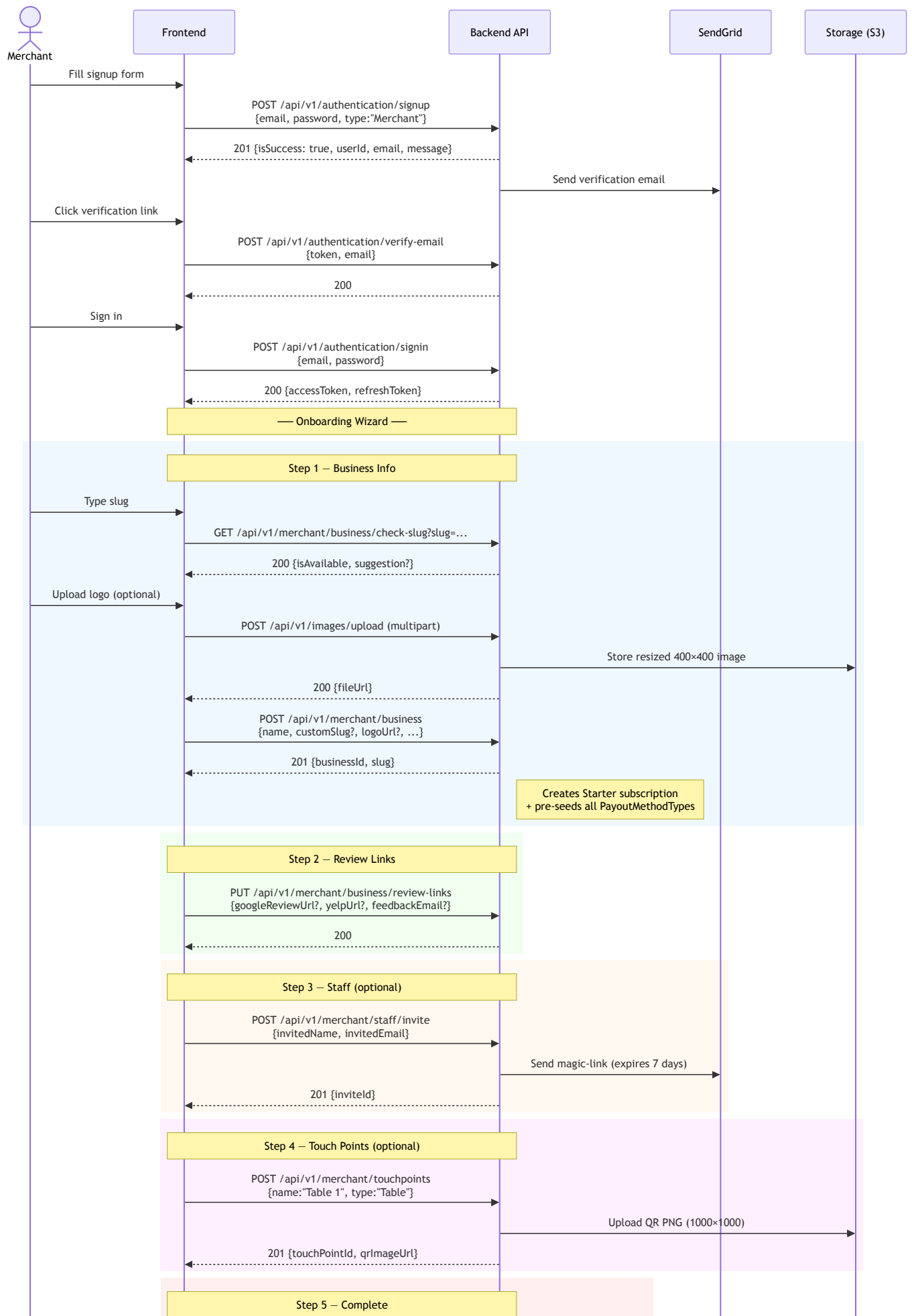
Status	ErrorCode	Description
200	—	Tip history returned
401	COMMON_UNAUTHORIZED	Missing or invalid token
404	STAFF_PROFILE_NOT_FOUND	No staff profile linked to this user

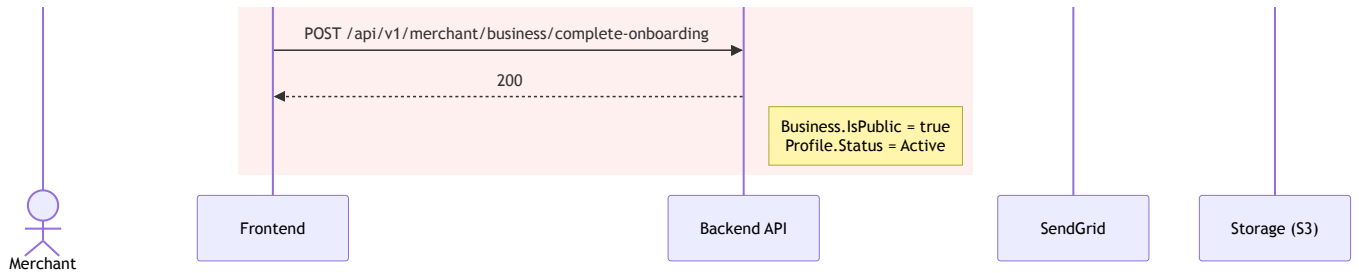
Part 2 — Business Flow Sequences

Each flow below has two representations:

- 1. **Sequence diagram** — shows all actors and API calls in order
- 2. **Step list** — exact endpoints, request bodies, and responses

Flow 1: Merchant Registration & Onboarding Wizard





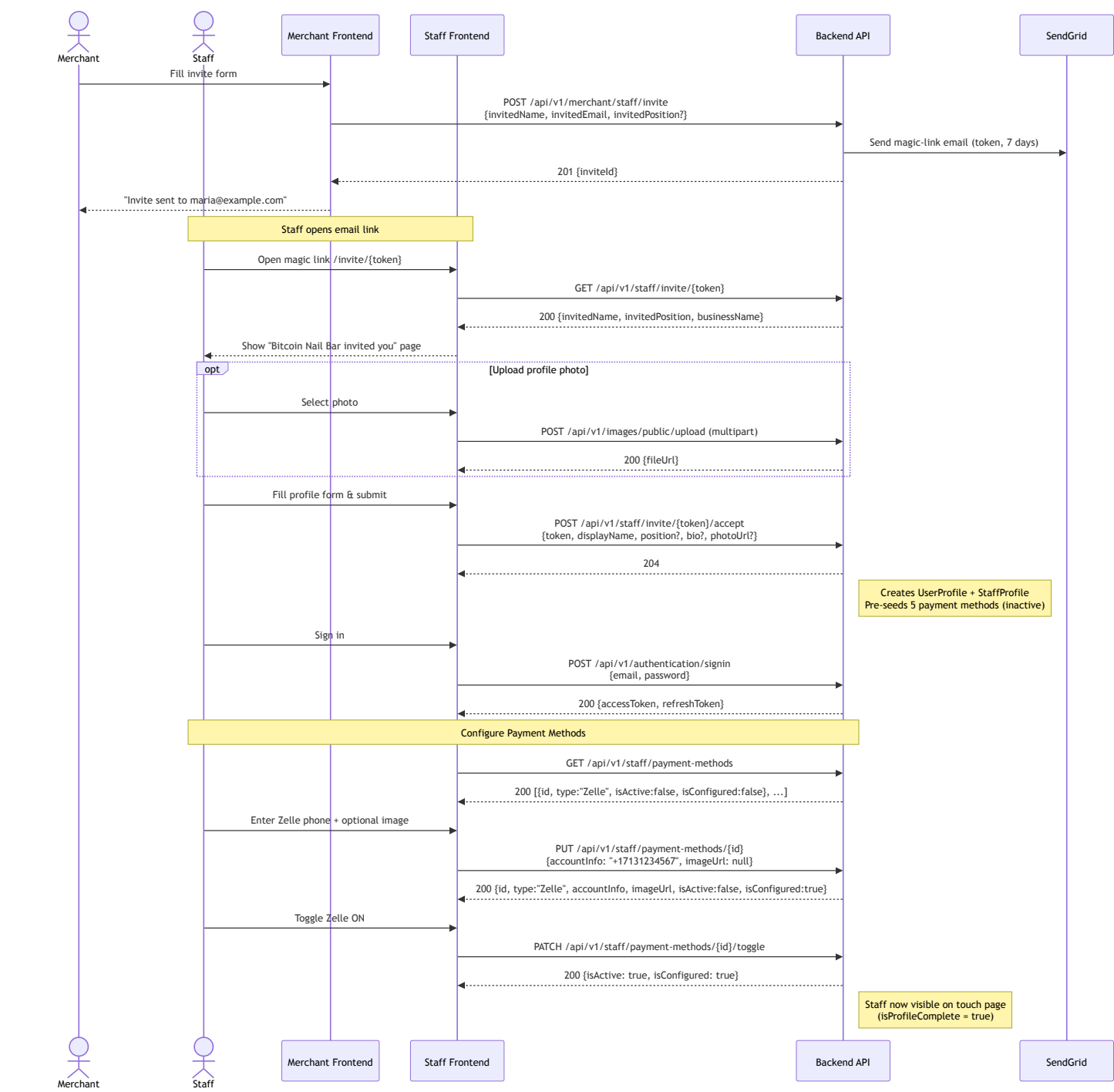
Step-by-step:

1. POST /api/v1/authentication/signup
{ email, password, confirmEmail: email, type: "Merchant", referralCode? }
→ 201 { isSuccess: true, userId, email, message }
2. POST /api/v1/authentication/verify-email
{ token (from email link), email }
→ 200
3. POST /api/v1/authentication/signin
{ email, password }
→ 200 { accessToken, refreshToken }

— ONBOARDING WIZARD —

- 4a. GET /api/v1/merchant/business/check-slug?slug=bitcoin-nail-bar
→ 200 { isAvailable: true } [called on each keystroke]
- 4b. POST /api/v1/images/upload [optional logo]
FormData: { file: <image> }
→ 200 { fileUrl }
- 4c. POST /api/v1/merchant/business [Step 1 – Business Info]
{ name, businessType?, address?, city?, state?, phone?,
timeZone?, description?, logoUrl?, customSlug? }
→ 201 { businessId, slug }
5. PUT /api/v1/merchant/business/review-links [Step 2]
{ googleReviewUrl?, yelpUrl?, facebookUrl?, feedbackEmail? }
→ 200
6. POST /api/v1/merchant/staff/invite [Step 3 – optional]
{ invitedName, invitedEmail, invitedPosition? }
→ 201 { inviteId }
7. POST /api/v1/merchant/touchpoints [Step 4 – optional]
{ name: "Table 1", type: "Table" }
→ 201 { touchPointId, qrImageUrl }
8. POST /api/v1/merchant/business/complete-onboarding [Step 5]
→ 200
(Business is now public; merchant profile → Active)

Flow 2: Invite Staff — New (Luồng A)



Step-by-step:

MERCHANT SIDE

1. POST /api/v1/merchant/staff/invite
{ invitedName: "Maria Garcia", invitedEmail: "maria@example.com",
invitedPosition: "Stylist" }
→ 201 { inviteId }
[API sends magic-link email – expires in 7 days]

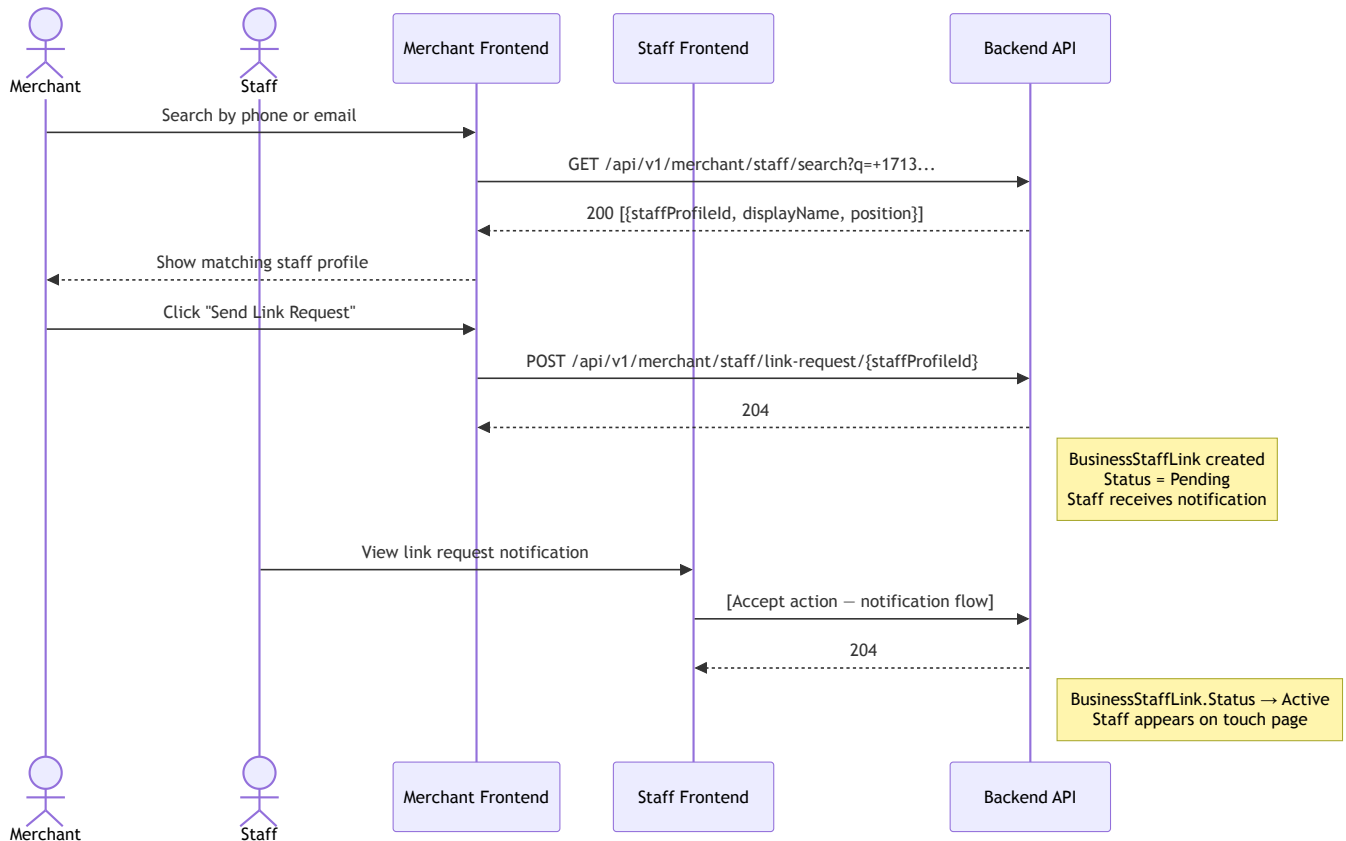
STAFF SIDE – opens magic link in browser

2. GET /api/v1/staff/invite/{token} (AllowAnonymous)
→ 200 { invitedName, invitedPosition, businessName }
3. POST /api/v1/images/public/upload [optional photo]
FormData: { file: <photo> }
→ 200 { fileUrl }
4. POST /api/v1/staff/invite/{token}/accept (AllowAnonymous)
{ token, displayName, position?, bio?, photoUrl? }
→ 204
[Creates UserProfile + StaffProfile + pre-seeds 5 payment methods, all inactive]
5. POST /api/v1/authentication/signin
{ email, password }
→ 200 { accessToken, refreshToken }

CONFIGURE PAYMENT METHODS

6. GET /api/v1/staff/payment-methods
→ 200 [{ id, type: "Zelle", isActive: false, isConfigured: false }, ...]
7. PUT /api/v1/staff/payment-methods/{id}
{ accountInfo: "+17131234567", imageUrl: null }
→ 200 { id, type: "Zelle", accountInfo: "+17131234567", imageUrl: null,
isActive: false, isConfigured: true }
8. PATCH /api/v1/staff/payment-methods/{id}/toggle
→ 200 { isActive: true, isConfigured: true }
[Staff is now visible on the customer touch page]

Flow 3: Link Existing Staff (Luồng B)



Step-by-step:

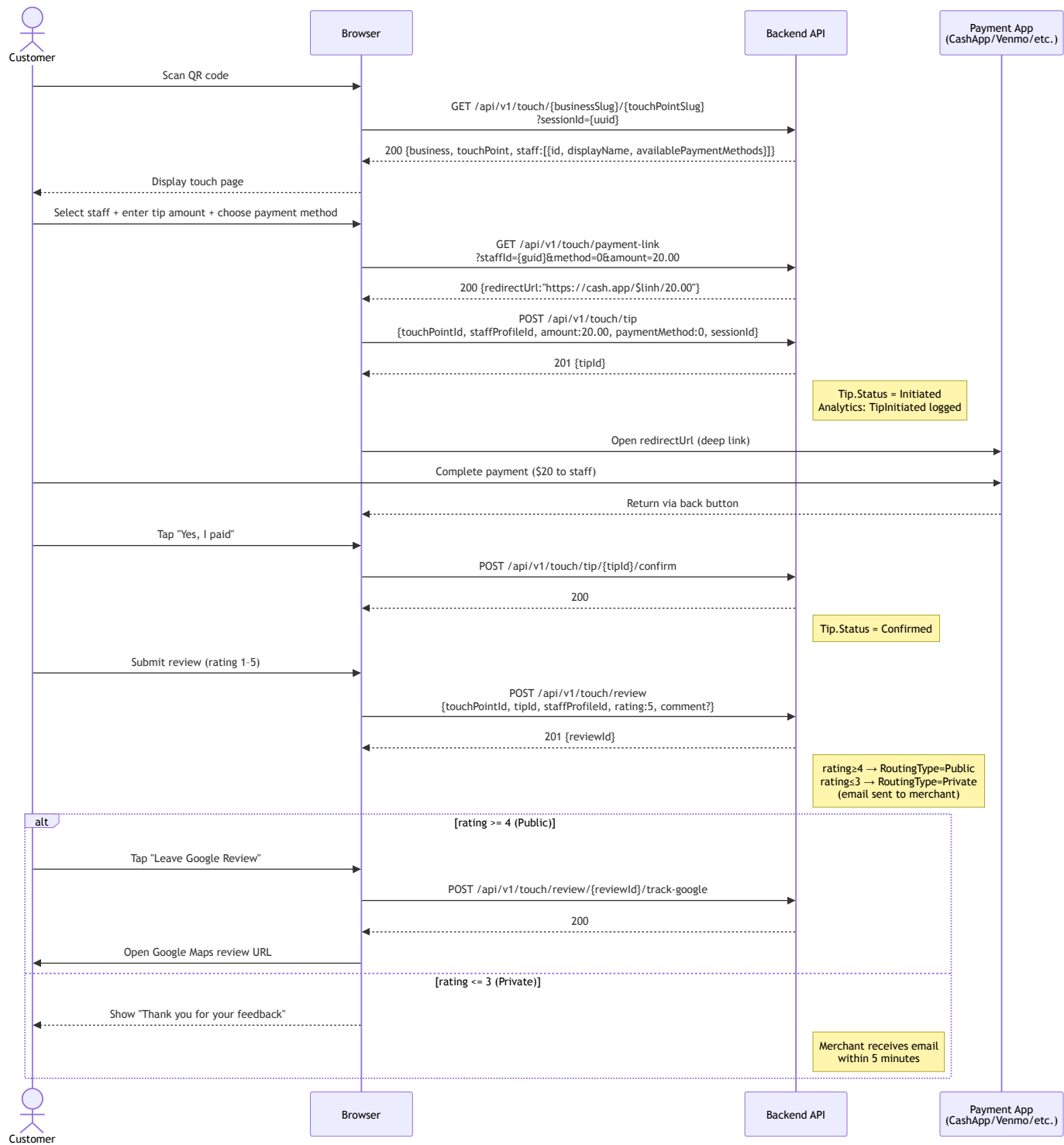
MERCHANT SIDE

1. GET /api/v1/merchant/staff/search?q=+17131234567
→ 200 [{ staffProfileId, displayName, position }]
2. POST /api/v1/merchant/staff/link-request/{staffProfileId}
→ 204
[Staff receives in-app notification to accept]

STAFF SIDE

3. Staff accepts link request via notification
→ BusinessStaffLink.Status → Active
[Staff becomes visible on touch page]

Flow 4: Customer Touch Flow (Single-Staff)



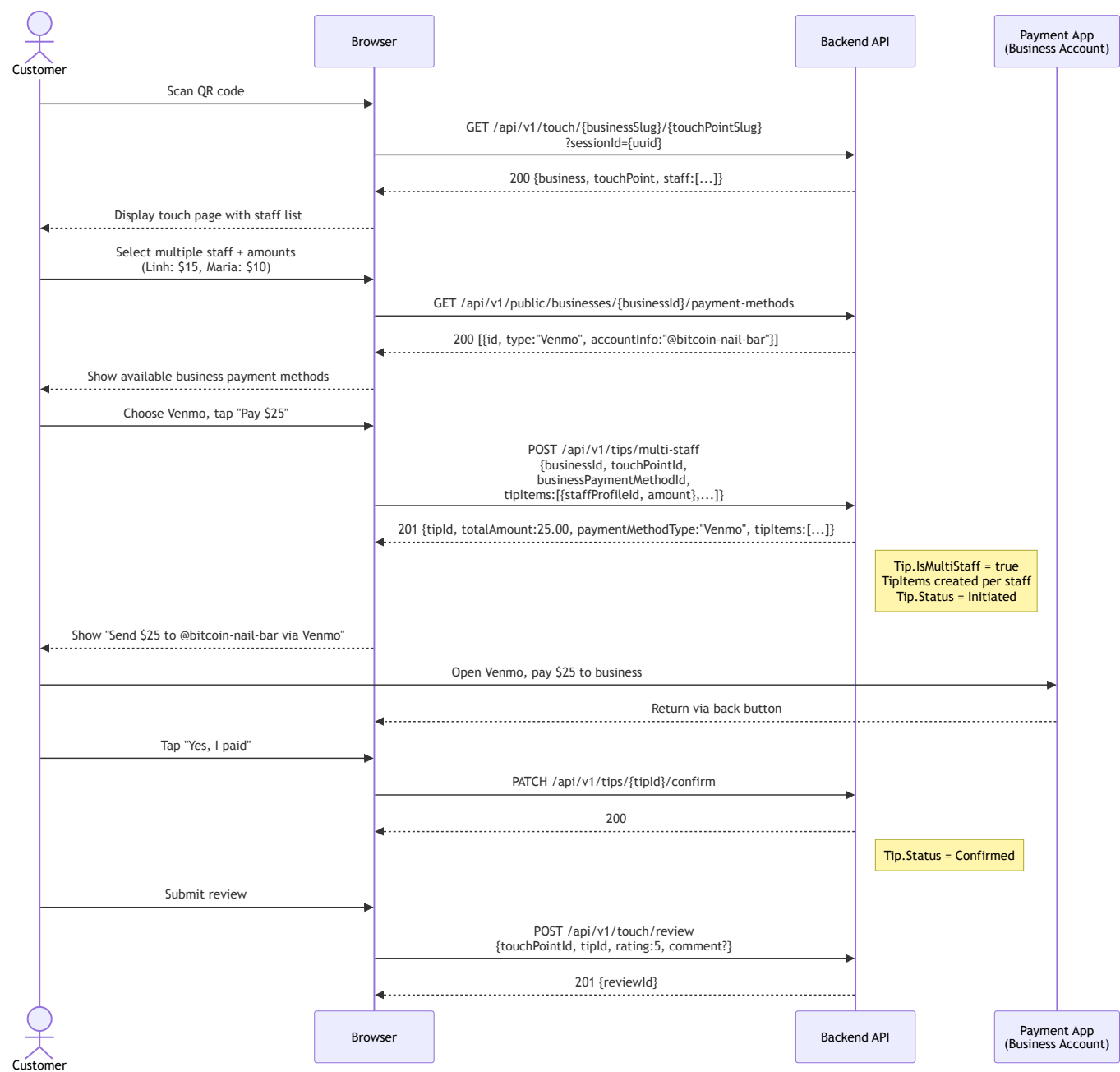
Step-by-step:

[Customer scans QR code – browser opens touch page]

1. GET /api/v1/touch/{businessSlug}/{touchPointSlug}?sessionId={uuid}
→ 200 { business, touchPoint, staff: [{ id, displayName, availablePaymentMethods }] }
2. [Customer selects staff member and tip amount]
3. GET /api/v1/touch/payment-link?staffId={guid}&method=0&amount=20.00
→ 200 { redirectUrl: "https://cash.app/\$linh/20.00" }
[Frontend opens redirectUrl in same tab]
4. POST /api/v1/touch/tip
{ touchPointId, staffProfileId, amount: 20.00, paymentMethod: 0, sessionId }
→ 201 { tipId } [Tip.Status = Initiated]
5. [Customer pays in CashApp/Venmo/etc., taps back to return]
6. POST /api/v1/touch/tip/{tipId}/confirm
→ 200 [Tip.Status = Confirmed]
7. POST /api/v1/touch/review
{ touchPointId, tipId, staffProfileId, rating: 5, comment?: "Amazing!" }
→ 201 { reviewId }
- 8a. [rating >= 4] POST /api/v1/touch/review/{reviewId}/track-google
→ 200 [customer clicked Google review button]
- 8b. [rating <= 3] Private feedback stored; email sent to merchant <5 min



Flow 4M: Customer Touch Flow (Multi-Staff)

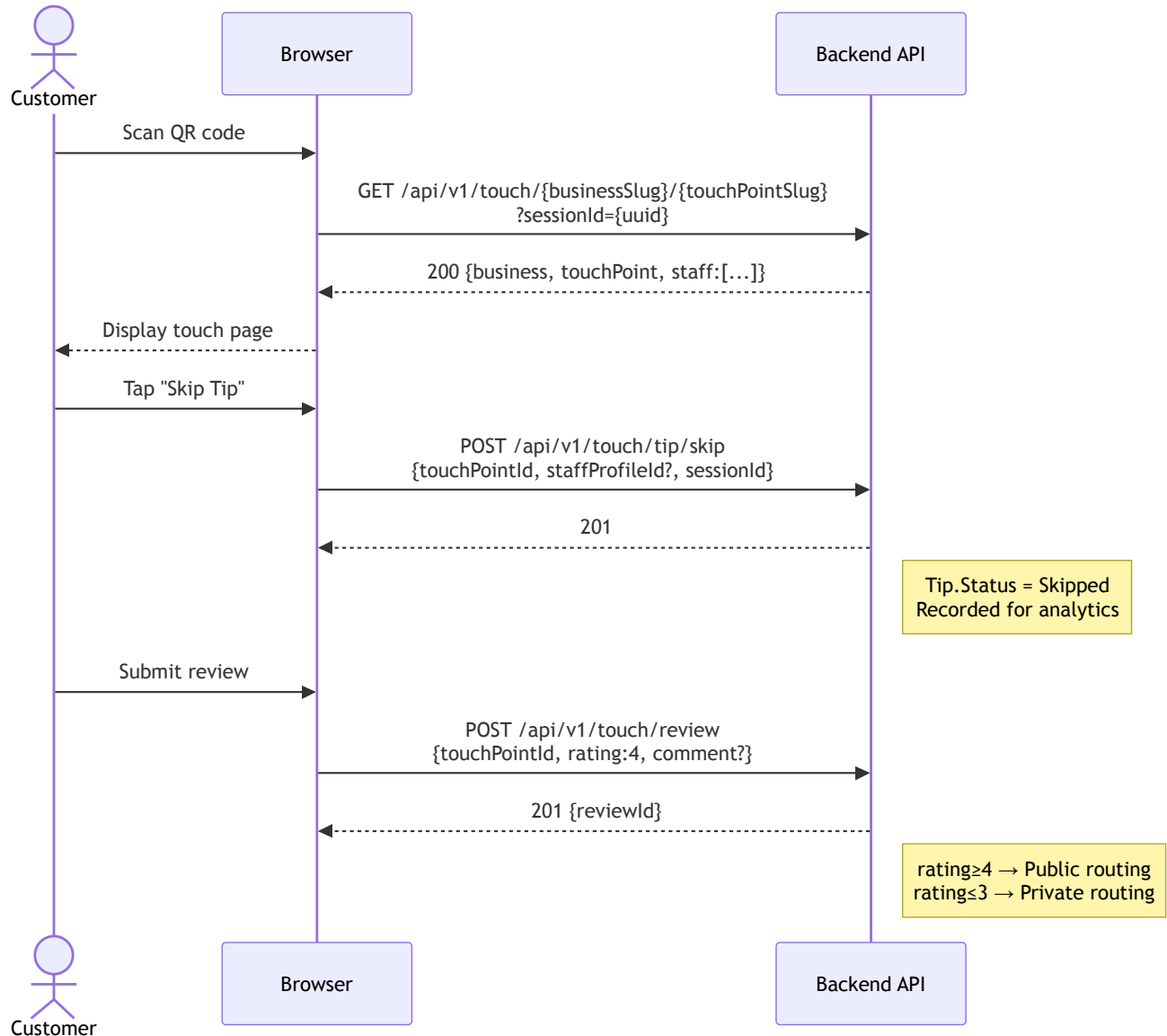


Step-by-step:

[Customer scans QR code – browser opens touch page]

1. GET /api/v1/touch/{businessSlug}/{touchPointSlug}?sessionId={uuid}
→ 200 { business, touchPoint, staff: [...] }
2. [Customer selects 2+ staff and allocates amounts]
3. GET /api/v1/public/businesses/{businessId}/payment-methods
→ 200 [{ id, type: "Venmo", accountInfo: "@bitcoin-nail-bar" }]
4. POST /api/v1/tips/multi-staff
{
 businessId,
 touchPointId,
 businessPaymentMethodId: "guid",
 tipItems: [
 { staffProfileId: "guid-linh", amount: 15.00 },
 { staffProfileId: "guid-maria", amount: 10.00 }
]
}
→ 201 { tipId, totalAmount: 25.00, paymentMethodType: "Venmo", tipItems: [...] }
5. [Frontend shows "Pay \$25 to @bitcoin-nail-bar via Venmo"]
[Customer pays in Venmo, returns via back button]
6. PATCH /api/v1/tips/{tipId}/confirm
→ 200 [Tip.Status = Confirmed]
7. POST /api/v1/touch/review
{ touchPointId, tipId, rating: 5, comment?: "Both were great!" }
→ 201 { reviewId }

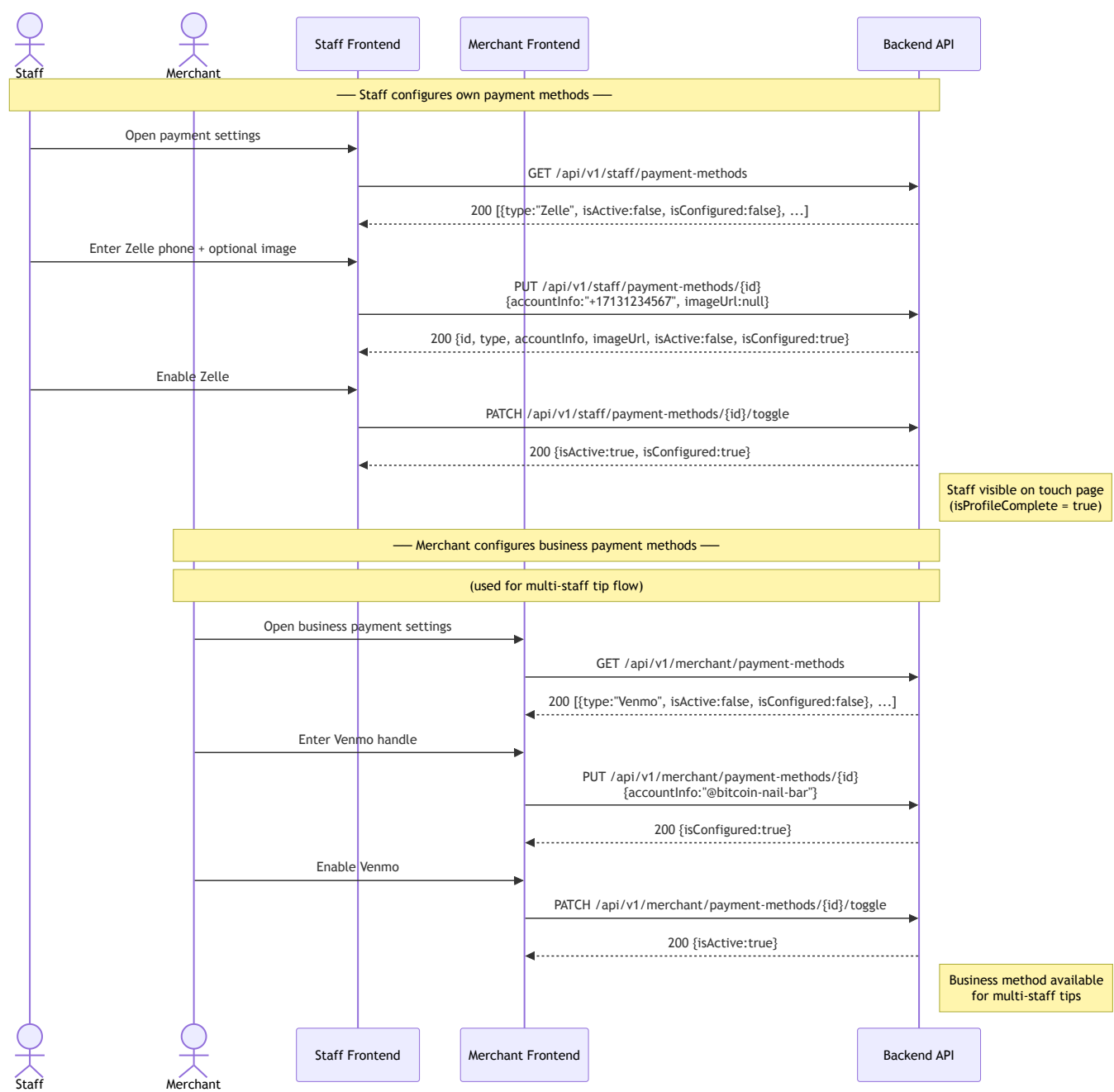
Flow 4b: Customer Skip Tip — Review Only



Step-by-step:

1. `GET /api/v1/touch/{businessSlug}/{touchPointSlug}?sessionId={uuid}`
→ `200 { ... }`
2. `POST /api/v1/touch/tip/skip`
`{ touchPointId, staffProfileId?, sessionId }`
→ `201` [Tip.Status = Skipped]
3. `POST /api/v1/touch/review`
`{ touchPointId, rating: 4, comment?: "Great service!" }`
→ `201 { reviewId }`

Flow 5: Payment Method Management



Step-by-step:

STAFF – Configure own payment methods

1. GET /api/v1/staff/payment-methods

→ 200 [{ id, type: "Zelle", isActive: false, isConfigured: false }, ...]

2. PUT /api/v1/staff/payment-methods/{id}

{ accountInfo: "+17131234567", imageUrl: null }

→ 200 { id, type: "Zelle", accountInfo: "+17131234567", imageUrl: null, isActive: false, isConfigured: true }

3. PATCH /api/v1/staff/payment-methods/{id}/toggle

→ 200 { isActive: true }

MERCHANT – Configure business payment methods (for multi-staff tips)

4. GET /api/v1/merchant/payment-methods

→ 200 [{ id, type: "Venmo", isActive: false, isConfigured: false }, ...]

5. PUT /api/v1/merchant/payment-methods/{id}

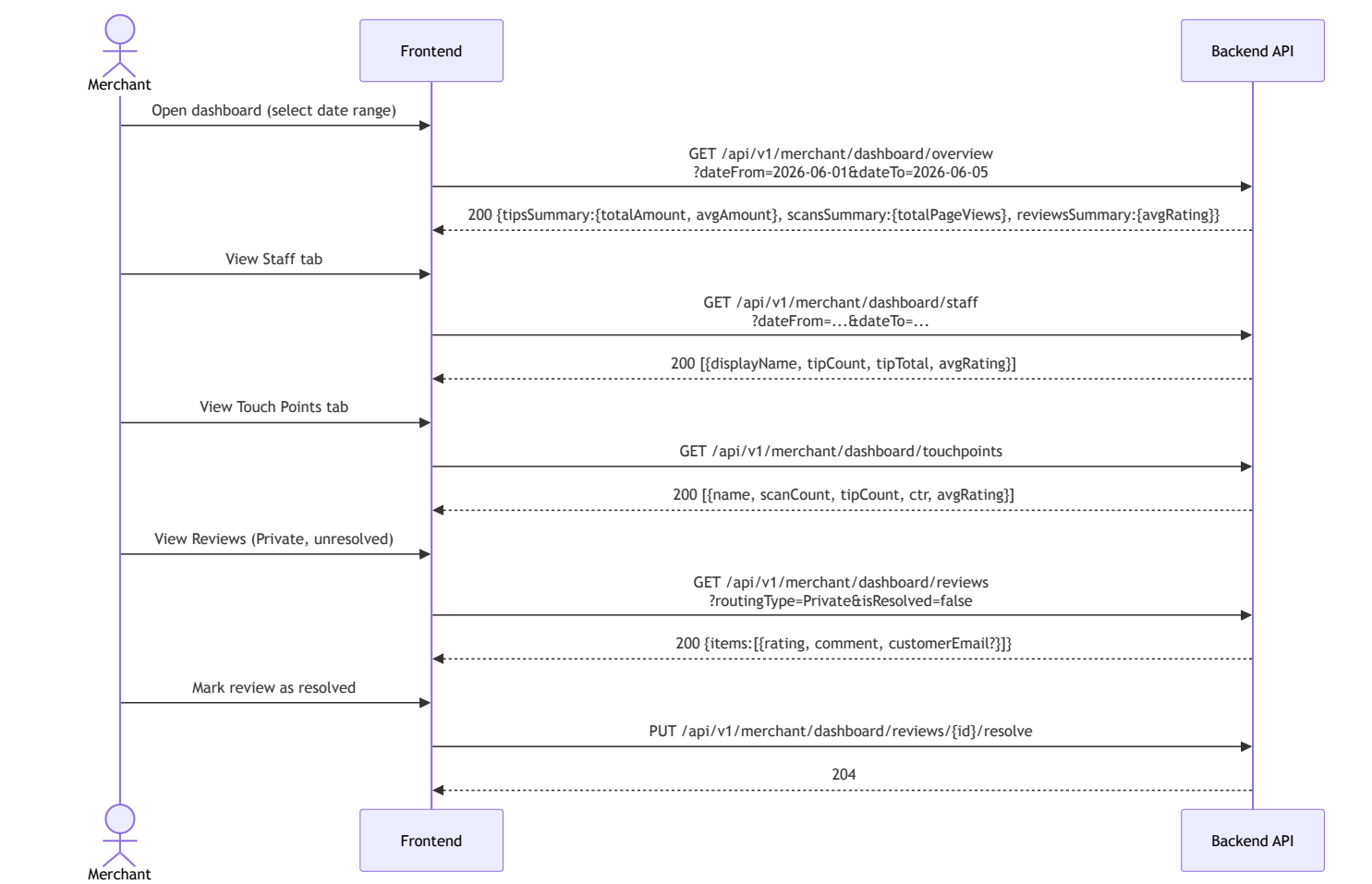
{ accountInfo: "@bitcoin-nail-bar" }

→ 200 { isConfigured: true }

6. PATCH /api/v1/merchant/payment-methods/{id}/toggle

→ 200 { isActive: true }

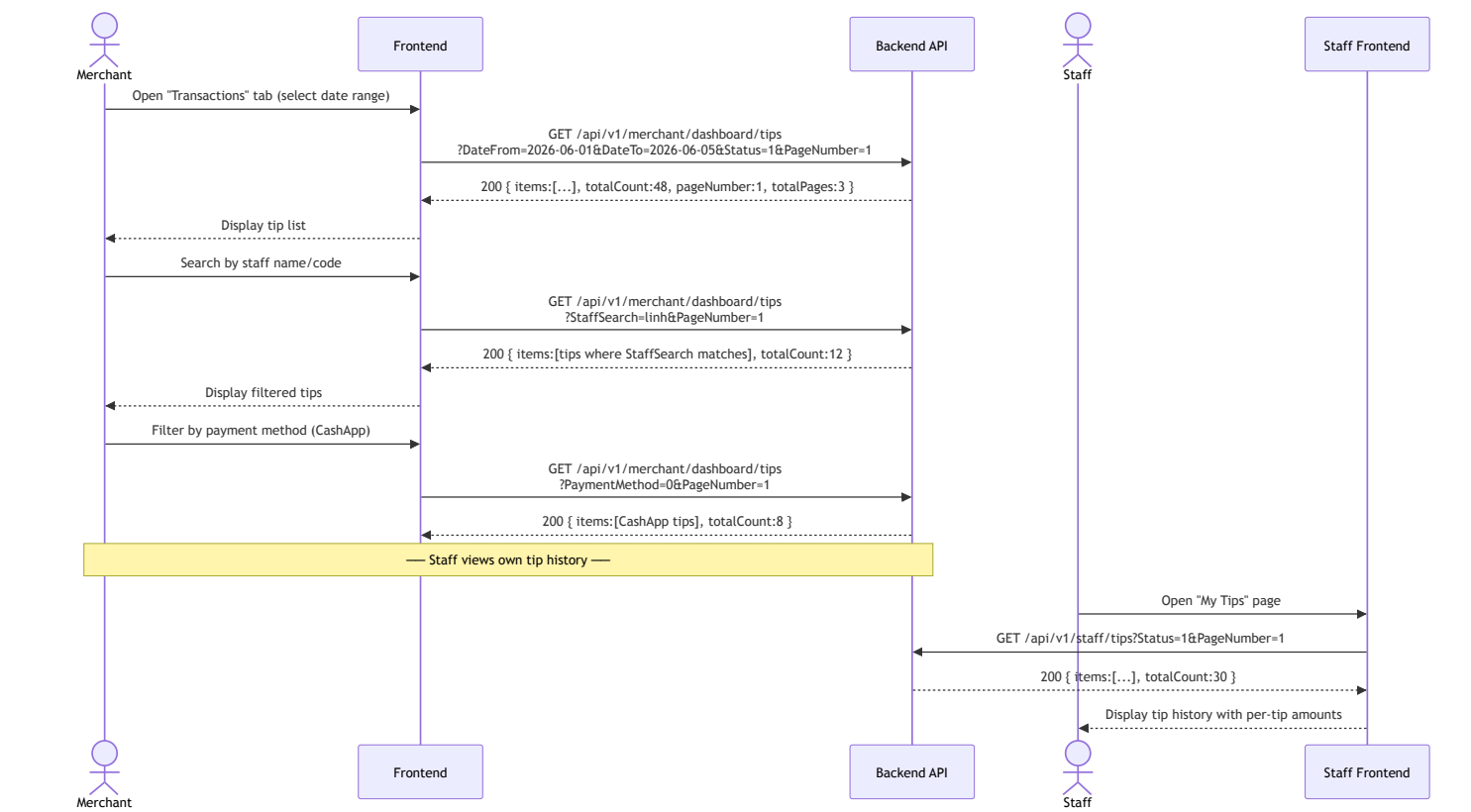
Flow 6: Dashboard Management



Step-by-step:

1. GET /api/v1/merchant/dashboard/overview?dateFrom=2026-06-01&dateTo=2026-06-05
→ 200 { tipsSummary: { totalAmount, avgAmount }, scansSummary: { totalPageViews, reviewsSummary: { totalCount, avgRating, ... } } }
2. GET /api/v1/merchant/dashboard/staff?dateFrom=...&dateTo=...
→ 200 [{ displayName, tipCount, tipTotal, avgRating, selectionCount }]
3. GET /api/v1/merchant/dashboard/touchpoints?dateFrom=...&dateTo=...
→ 200 [{ name, scanCount, tipCount, ctr, avgRating }]
4. GET /api/v1/merchant/dashboard/reviews?routingType=Private&isResolved=false
→ 200 { items: [{ rating, comment, customerEmail? }] }
5. PUT /api/v1/merchant/dashboard/reviews/{id}/resolve
→ 204

Flow 8: Tip Transaction History



Step-by-step:

MERCHANT – View tip transactions

1. GET /api/v1/merchant/dashboard/tips

?DateFrom=2026-06-01&DateTo=2026-06-05&Status=1&PageNumber=1&PageSize=20

→ 200 { items: [...], totalCount: 48, pageNumber: 1, totalPages: 3 }

2. [Merchant filters by staff name]

GET /api/v1/merchant/dashboard/tips?StaffSearch=linh&PageNumber=1

→ 200 { items: [tips containing "linh" in DisplayName, StaffCode, Email, or Phone], ..

3. [Merchant filters by exact staff ID]

GET /api/v1/merchant/dashboard/tips?StaffProfileId={guid}&PageNumber=1

→ 200 { items: [all tips – single and multi-staff – for that staff], ... }

4. [Merchant filters multi-staff tips only]

GET /api/v1/merchant/dashboard/tips?IsMultiStaff=true&PageNumber=1

→ 200 { items: [multi-staff tips with populated tipItems array], ... }

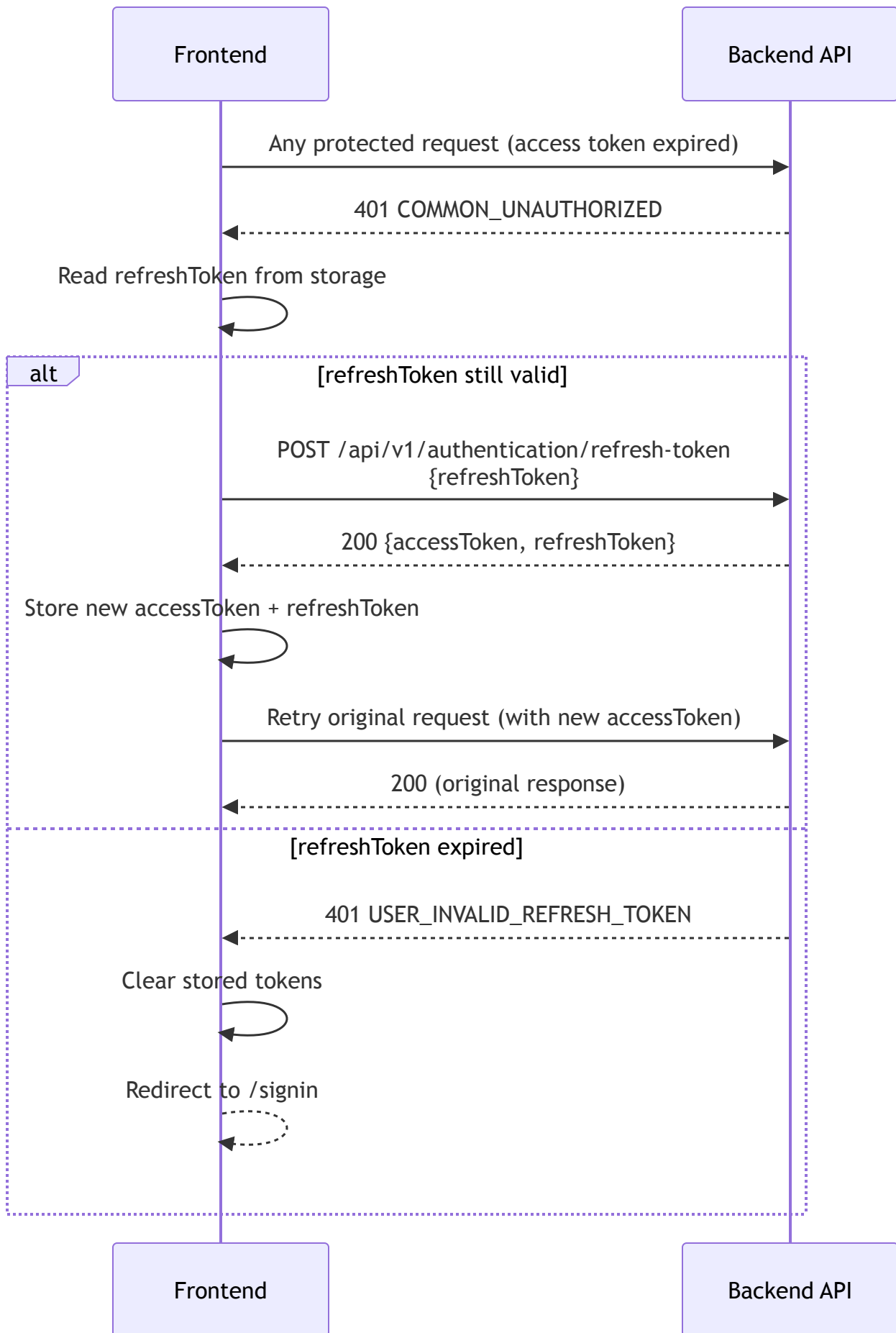
STAFF – View own tip history

5. GET /api/v1/staff/tips?Status=1&PageNumber=1

→ 200 { items: [...], totalCount: 30 }

[Multi-staff tips show amount = staff's portion, totalAmount = full tip]

Flow 7: Auto Refresh Token



Step-by-step:

[Access token expires – 401 received from any protected endpoint]

1. POST /api/v1/authentication/refresh-token
{ refreshToken: "<stored>" }
→ 200 { accessToken: "newToken", refreshToken: "newRefresh" }
2. Store new tokens; retry original request with new accessToken.

[If refreshToken is also expired]

- 401 USER_INVALID_REFRESH_TOKEN
→ Clear tokens; redirect to /signin

Part 3 — Enums Reference

PaymentMethod (customer → staff payment method)

Used in POST /api/v1/touch/tip and GET /api/v1/touch/payment-link .

Value	Name	Description
0	CashApp	Cash App \$cashtag
1	Venmo	Venmo handle
2	Zelle	Zelle (phone or email, no redirect)
3	PayPal	PayPal.me link
4	Other	Skipped tip (internal only)
5	AppleCash	Apple Cash (phone, no redirect)

PayoutMethodType (staff/business payment method type)

Used in StaffPaymentMethodDto.Type and BusinessPaymentMethodDto.Type .

Value	Name
1	Zelle
2	BankWire
3	PayPal
4	Venmo
5	CashApp
6	AppleCash
7	VlinkPay

TouchPointType

Value	Name	Description
0	Table	Shows all active staff
1	FrontDesk	Shows all active staff
2	Receipt	Shows all active staff
3	StaffCard	Shows only the assigned staff member

TipStatus

Value	Name	Description
0	Initiated	Customer clicked Pay; payment not confirmed
1	Confirmed	Customer confirmed payment
2	Skipped	Customer chose to skip tip

ReviewRoutingType

Value	Name	Threshold
0	Public	Rating 4–5 → show Google/Yelp buttons
1	Private	Rating 1–3 → collect private feedback + email merchant
2	Skipped	Customer skipped review

BusinessStaffLinkStatus

Value	Name	Description
0	Pending	Invite sent, not yet accepted
1	Active	Staff linked and visible on touch page
2	Inactive	Hidden from touch page, data preserved
3	Rejected	Staff declined the link request

StaffInviteStatus

Value	Name
0	Pending
1	Accepted
2	Expired
3	Cancelled

Part 4 — Standard Error Response Format

All 4xx and 5xx responses follow this shape:

```
{
  "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
  "title": "Bad Request",
  "status": 400,
  "errors": {
    "amount": ["TIP_AMOUNT_TOO_LOW"]
  },
  "errorCode": "COMMON_VALIDATION_ERROR"
}
```

Single-error response (e.g., 404):

```
{
  "type": "...",
  "title": "Not Found",
  "status": 404,
  "errorCode": "BUSINESS_NOT_FOUND"
}
```

Validation error (FluentValidation):

```
{
  "status": 400,
  "errorCode": "COMMON_VALIDATION_ERROR",
  "errors": {
    "name": ["BUSINESS_NAME_REQUIRED"],
    "amount": ["TIP_AMOUNT_TOO_LOW", "TIP_AMOUNT_TOO_HIGH"]
  }
}
```

Rate limit exceeded (429):

```
{
  "status": 429,
  "errorCode": "COMMON_RATE_LIMIT_EXCEEDED",
  "retryAfter": 60
}
```

Common Error Codes

Code	HTTP Status	When
COMMON_VALIDATION_ERROR	400	FluentValidation failure
COMMON_NOT_FOUND	404	Generic not found
COMMON_UNAUTHORIZED	401	No/invalid token
COMMON_FORBIDDEN	403	Authorized but not permitted
COMMON_RATE_LIMIT_EXCEEDED	429	Rate limit hit
COMMON_INTERNAL_SERVER_ERROR	500	Unexpected server error